A Physics-Informed Neural Network (PINN) framework for generic bioreactor modelling

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Abstract

Many previous studies have explored hybrid semiparametric models merging Artificial Neural Networks (ANNs) with mechanistic models for bioprocess applications. More recently, Physics-Informed Neural Networks (PINNs) have emerged as promising alternatives. Both approaches seek to incorporate prior knowledge in ANN models, thereby decreasing data dependency whilst improving model transparency and generalization capacity. In the case of hybrid semiparametric modelling, the mechanistic equations are hard coded directly into the model structure in interaction with the ANN. In the case of PINNs, the same mechanistic equations must be "learned" by the ANN structure during the training. This study evaluates a dual-ANN PINN for generic bioreactor problems illustrated with 2 case studies. Furthermore, the proposed PINN is benchmarked against the general hybrid semiparametric bioreactor model under comparable prior knowledge scenarios. Our findings show that PINNs can level the prediction accuracy of hybrid semiparametric models for simple problems. PINNs

exhibit significant performance degradation in extended temporal extrapolation or complex problems involving high-dimensional process states subject to time-varying control inputs. The latter is mainly related to the more complex multi-objective training of PINN structures.

Keywords

Bioprocess modelling; Physics-informed neural networks; Hybrid semiparametric modelling; Deep artificial neural networks; ADAM algorithm; Fed-batch bioreactor

1. Introduction

Due to the intricate mechanistic complexity of cell culture systems, developing predictive models for optimization and control at an acceptable cost remains a challenge in the bioprocessing industries Hong et al. (2021). Traditional modelling approaches involve mathematical equations derived from first principles, including mass and energy balances, reaction kinetics, and transport phenomena. While these models can elucidate numerous aspects of process behaviour, their development is time-consuming and requires extensive domain knowledge. Furthermore, these models are inherently limited in their capacity to represent the global characteristics of most biological systems in specific application scenarios, and the efficiency of their solution is significantly lower than theoretically possible (Jarka Glassey et al., 2011; Van Impe et al., 2013).

In recent years, data-driven modelling has received significant attention leveraging on technical progress in data acquisition and process analytical technology. Particularly deep artificial neural networks (deep ANNs) are increasingly considered for bioprocess modelling due to their ability to infer intricate patterns from data with minimal domain knowledge requirements (Helleckes et al., 2023). However, their dependency on high-quantity and high-quality data, which are often scarce, and their limited generalization capability constitute major drawbacks (Thomas Kipf et al., 2018). To mitigate these limitations, hybrid modelling approaches have emerged that combine ANNs and mechanistic modelling in a common workflow. The most frequently reported approach is hybrid semiparametric modelling, wherein ANNs and mechanistic equations are interlinked in a common model structure, thereby enhancing the model's accuracy and transparency (Psichogios and Ungar 1992; Thompson and Kramer, 1994; Schubert et al.,1994; Oliveira, 2004; Kadlec et al.,2009; Von Stosch et al., 2014). Numerous application examples have demonstrated the effectiveness of hybrid semiparametric modelling for bioreactor systems (e.g., Narayanan et al., 2023; Bayer et al., 2023; Pinto et al., 2023; Ramos et al., 2024). A common path involves using intensified Statistical Design of Experiments (iDoE) in combination with hybrid modelling to dynamically infer the impact of critical process parameters (CPP) on critical quality attributes (CQA) (Von Stosch et al., 2014). These methods not only help optimize process conditions but also reduce experimental runs, thus saving both time and resources. As a result, process engineers can achieve more reliable, scalable, and precise outcomes thereby enhancing overall productivity in industrial applications (Smiatek et al., 2020).

A recent addition to bioprocess modelling is Physics-Informed Neural Networks (PINNs). PINNs incorporate a system's known governing physical law, usually described by partial differential equations (PDEs) or ordinary differential equations (ODEs), into the training process of a deep neural network (DNN) (Raissi et al., 2019). The training minimizes two loss functions simultaneously, a data loss function based on process observations, and a physics loss function derived from physics equations. Unlike conventional DNNs, PINNs can handle complex problems, where precise data collection is difficult due to real-world measurement limitations. PINNs and hybrid semiparametric models are thus related concepts with similar goals, i.e. the integration of prior process knowledge in ANN models thereby reducing data dependency. Whereas in semiparametric modelling the physics equations are hard coded directly in the model structure, in the case of PINNs the same physics equations are embedded in a physics loss function that is minimized during the training. The general principle of a PINN was already postulated in the early study by Thomson and Kramer (1994) who categorized hybrid modelling as design methods, where prior knowledge is hard coded in the model structure, and training methods where, prior knowledge is embedded in the training loss function. Since the pioneering study by Raissi et al. (2019), PINNs are now taking the first steps in the bioprocess modelling field. Due to their similarities, hybrid semiparametric modelling and PINNs are sometimes interchangeably referred to in the literature (e.g., Bangi et al., 2022; Cui et al., 2024). Adebar et al. (2024) recently proposed a PINN approach for mammalian cultivations using truncated Taylor series expansions to approximate growth kinetics. Simplifications such as pseudo-first-order kinetics and decoupled individual process outcomes were introduced. Despite the simplifications, they successfully modelled with reasonable accuracy the dynamics of key variables such as viable cell density, glucose, lactate, and product. In a different study, Li et al. (2024) developed a complex PINNs framework using the multi-stage Koopman method for microbial growth modelling. The Koopman method served to map the process dynamics into a high-dimensional linear space and to model each growth stage separately in the linear space. Yang et al. (2024) compared parallel hybrid modelling, serial hybrid modelling, and PINNs in a pilot fed-batch CHO culture. They applied the multiple-shooting method to divide the culture into 24-hour intervals with constant feed. Their results suggested that PINNs have superior predictive power than the widely adopted semiparametric approaches. Moayedi et al. (2024) compared a Recurrent Neural Network (RNN) with a recurrent PINN in a simulation case study with 2 continuous reactors connected in series. They used the Euler method to approximate the derivatives of state variables to evaluate the physics loss function. They concluded that PINN was superior to conventional RNN. Jul-Rasmussen et al. (2025) recently presented a comparative study of hybrid semiparametric modelling and physics-informed recurrent neural networks (PIRNNs) for a pilot-scale bubble column aeration process. The study compared each approach in terms of training ease, adherence to governing equations, prediction accuracy with less frequent measurements, and performance with limited data. Their study found that hybrid semiparametric modelling outperformed the recurrent PINN approach for the pilot-scale bubble column aeration process.

PINNs have gained significant traction in scientific computing, but their advantages and development opportunities for bioprocess applications need more evidence. Typical problems arising with PINNs are the choice of the ANN structure, handling time-varying control inputs, enforcing initial condition (IC), and convergence of the multi-objective loss function, which combines physics, data, and IC constraints. One critical issue in PINN training is the trade-off between data loss and physics loss, which directly influences model convergence. In this study, we propose a dual-ANN PINN structure that is generally applicable to stirred-tank bioreactor problems under time-varying and batch-varying control inputs. This structure is evaluated in 2 case studies and compared with the general bioreactor hybrid model (Oliveira, 2004; Pinto et al., 2022). The remainder of this paper is organized as follows: section 2 details the methodological framework, including the PINN structure and training strategies. Section 3 introduces the case studies, encompassing both simple and complex bioreactor problems. Section 4 presents the modelling results and comparative performance analysis. Section 5 provides an in-depth discussion of the findings in the context of current literature. Finally, Section 6 presents the main conclusions.

2. Methods

2.1. PINN structure for bioreactor systems

The state-space equation of a perfectly mixed stirred tank bioreactor takes the following general form,

$$\frac{dC}{dt} = Sr(C,u) - DC + DC_{in},\ C(0) = C_0 \qquad \text{(1a)},$$

Where C is a $(n \times 1)$ state vector of concentrations of relevant biochemical species, S is a $(n \times m)$ matrix of yield coefficients, $r(C,u)$ is a $(m \times 1)$ vector of biological reaction kinetics, $D = F/V$ is the dilution rate (for simplicity we consider the liquid volume, V, to be controlled by a single feed stream with flow rate F), C_{in} is a $(n \times 1)$ vector of concentrations in the feed stream and t the independent variable time. The overall material balance equation for constant liquid density and a single feed stream is defined as,

$$\frac{dV}{dt} = F,\ V(0) = V_0 \qquad \text{(1b)}$$

Based on the bioreactor state-space Eqs. (1a, b), a PINN structure is proposed assuming that the reaction kinetics term is unknown, i.e., there are no defining equations for the term $r(C,u)$ (Figure 1A). The PINN structure is composed of a state parameterization feed-forward neural network (FFNN-S) and a reaction kinetics feed-forward neural network (FFNN-R). The rationale behind this dual-FFNN structure is uncoupling the parameterization of dynamic state variables over time and reaction kinetics over state variables, thereby improving the convergence stability of the multi-objective training processes as further discussed below.

Both network structures are multi-layered FFNNs with an input layer, one or more hidden layers (to capture the complex interactions and nonlinearities in the system), and an output layer:

$$y = \sigma\left(W_L . \sigma\left(W_{L-1} . \sigma\left(W_1 . x + b_1\right) + b_{L-1}\right) + b_L\right) \quad (2),$$

with y a vector of outputs, x a vector of inputs, W_i a matrix of weights of the ith layer, b_i a vector of bias parameters and σ the activation function.

Following the structure of Eq. (2), the FFNN-S is implemented as a one-step predictor. It receives as inputs the state variables, $C(t-\delta t)$, the control inputs $u_{[t-\delta t,t]}$ (which are constant over time intervals $[t-\delta t,t]$), and time intervals, δt. When the time step δt is constant it may drop from the FFNN-S inputs. Of note that δt might not be constant due to heterogeneous data sparsity. The predicted concentrations are computed from the neural network outputs as follows:

$$C(t) = C(t-\delta t) + y(t)\delta t \quad (3)$$

Eq. (3) ensures that $C(t) = C(t-\delta t)$ when $\delta t = 0$, which eliminates the need for initial boundary conditions in the loss function used to train the PINN. Automatic differentiation (AD) is applied to compute dC/dt, which is required in the physics loss function (further described below). The second neural network, FFNN-R, computes the reaction kinetics at the current time, $r(t) = y(t)$ as a function of concentrations at the current time, $C(t)$, and of control inputs at the current time, $u(t)$. The outputs of the PINN are thus $C(t)$, $r(t)$ and dC/dt. The outputs of the PINN serve to calculate data loss and physics loss. The prior knowledge of the material balance equation is not embedded directly in the model structure but is rather used to compute the physics loss, as further discussed below.

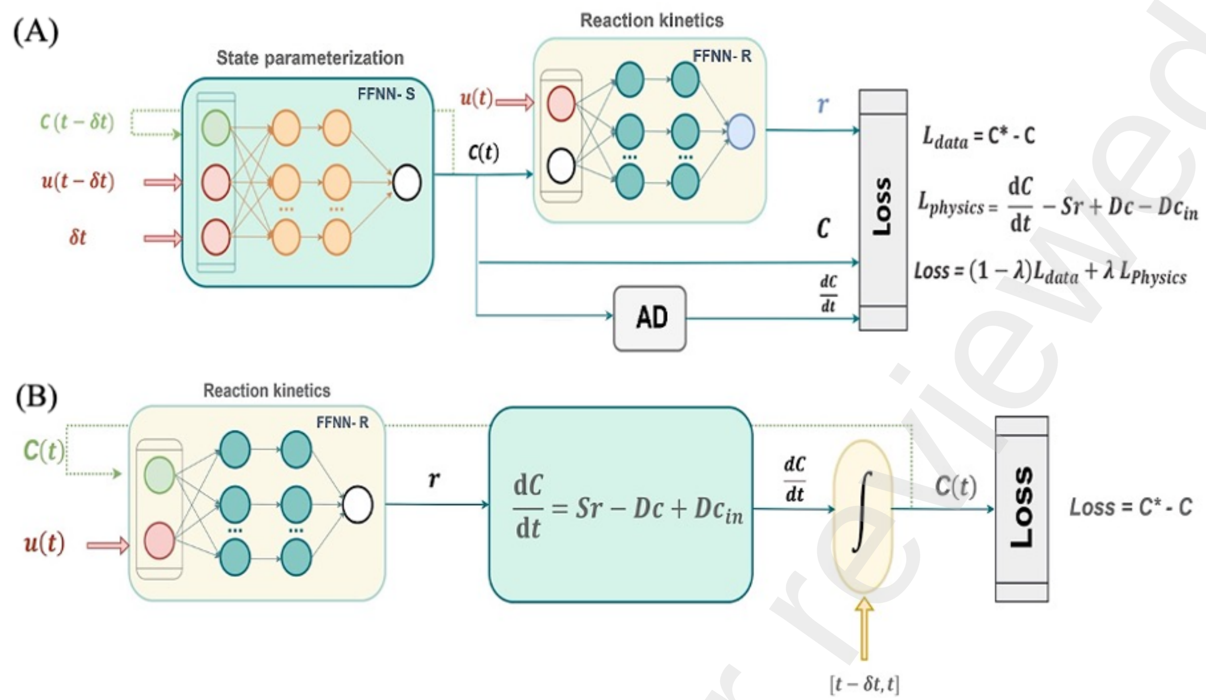


Figure 1: Schematic diagram of hybrid model structures for a generic stirred-tank bioreactor system. **(A)** Dual-FFNN PINN structure where physical equations are embedded in the training loss function, **(B)** Hybrid semiparametric structure where physical equations are hard-coded directly in the model structure.

2.2. Hybrid semiparametric model for bioreactor systems

This study compares the dual-FFNN PINN structure with the more traditional hybrid semiparametric approach. Here we follow the previously published general bioreactor hybrid model concept (Oliveira, 2004; Pinto et al., 2022) applied in many different studies. The hybrid semiparametric model Figure (1B) contains a single neural network that models the reaction kinetics as a function of the concentrations of species and control inputs. This network parallels the FFNN-R of the PINN structure. The calculated reaction kinetics pass to a system of ODEs derived from the material balance equations. This is a fundamental difference to the PINN structure, as the prior knowledge ODEs are hard-coded directly in the model structure.

The ODEs are numerically integrated over time intervals $[t-\delta t,t]$ from initial conditions, $C(t-dt)$ to the current concentrations, $C(t)$. The integration was done numerically with a Runge-Kutta 4th order method (RK4). This is another fundamental difference to PINNs, as the hybrid semiparametric approach requires integration whereas the PINN approach requires differentiation. However, it is important to note that the discretization of the process to match sampling time steps can impact the derivative approximation and overall solution accuracy, as highlighted by Cruz-Bournazou et al. (2022). Comparing computed and measured concentrations results in a data loss that is minimized during the training (further described below).

2.3. Training methods

To ensure comparability, the training, testing, and validation methods were identical for both modelling approaches, except for the calculation of the loss function which is intrinsically different in both approaches. The standard Adaptive Moment Estimation Method (ADAM) was applied to minimize the total loss. Specifically, the standard ADAM method described in Kingma and Ba, (2014) was applied. In the case of the PINN structure, a loss function with two terms is minimized. The data loss term is computed as the mean sum of squares as follows,

$$\mathcal{L}_{data}=\frac{1}{N\times M}\sum_{i=1}^{N}\sum_{j=1}^{M}\left(\frac{C_{i,j}^{*}-C_{i,j}}{\sigma_{Cj}}\right)^{2} \quad (4),$$

With N the number of observations, M the number of biochemical species, $C_{i,j}^{*}$ and $C_{i,j}$ the measured and model predicted concentrations respectively of biochemical species j at time instant i, and σ_{Cj} the standard deviation of the concentration of biochemical species j over the set of observations. The physics loss is computed as follows,

$$e=\frac{dC}{dt}-Sr+DC-DC_{in} \quad (5a),$$

$$\mathcal{L}_{physics} = \frac{1}{K \times M} \quad \sum_{i=1}^{K} \sum_{j=1}^{M} \left(\frac{\delta t \, e_{i,j}}{\sigma_{Cj}} \right)^2 \qquad (5b),$$

with K representing the number of collocation points and $e_{i,j}$ denoting the physics residual of biochemical species j at collocation point i. The number of collocation points does not coincide with the number of observations. A random sampling collocation method was employed to distribute these points across the domain space. Furthermore, it is ensured that the total number of residuals, $(N + K)\, M$, is always higher than the total number of weights of the dual-FFNN PINN structure. Finally, the total loss is computed as a weighted sum of the two loss terms:

$$\mathcal{L}_{total} = (1 - l)\mathcal{L}_{data} + l\, \mathcal{L}_{physics} \qquad (6)$$

The parameter l is a user defined parameter between 0-1 used to adjust the relative importance of the physics and data loss terms. For the hybrid semiparametric structures, the physics loss function drops, thus the total loss is simply given as $\mathcal{L}_{total} = \mathcal{L}_{data}$. For comparability, the data was partitioned consistently for both the PINN and semiparametric structures. Specifically, the dataset was divided into training, validation, and testing subsets, with the specific partitioning varying across case studies (further described in Section 3). To mitigate overfitting, either cross-validation or stochastic regularization were applied. Based on the study by Pinto et al. (2022), a minibatch size stochastic regularization scheme was applied.

The computation of gradients is a critical step for the success of both models. In the hybrid semiparametric case, sensitivity equations are often applied (Oliveira, 2004; Pinto et al., 2022). However, for comparability, the automatic differentiation method implemented in the python package PyTorch Baydin et al. (2018) was applied in both cases. All methods described here were implemented in

Python. The results reported in this study were obtained using Windows 11 Pro on a PC with an Intel Core i9 CPU with 32 GB RAM.

2.4. Model discrimination

The selection of the best model was based on test error, considering only the data loss term. The model with the lowest data loss on the test partition was chosen as the optimal one. Additionally, a probabilistic approach was applied using a modified Akaike Information Criterion with second-order bias correction (AICc). In this case, the total loss from Eq. (7) was considered, incorporating both the training data residuals and physics residuals to evaluate model performance. The inclusion of physics residuals is necessary because PINNs are often trained with a number of data residuals lower than the number of weights.

$$AICc = (N+K) \times M\, ln(\mathcal{L}_{total}) + 2\, nw + \frac{2\, nw\, (nw+1)}{(N+K) \times M - nw - 1} \quad (7)$$

The AICc penalizes the model complexity (i.e., the total number of network parameters, nw), ensuring that models with excessive complexity are not favored unless they provide a significantly better fit. The model with the lowest AICc score was selected as the best model.

3. Case studies

3.1. Case study 1 - Logistic biomass growth in a fed-batch bioreactor

Case study 1 is a very simple logistic growth process in a fed-batch bioreactor with only 2 state variables described by two ODEs Eqs. (8 a, b) and the logistic growth model Eq. (8c),

$$\frac{dX}{dt} = \mu X - DX \quad (8a),$$

$$\frac{dV}{dt} = F \quad (8b),$$

$$\mu = \mu_{max}\left(1 - \frac{X}{X_{max}}\right) \quad (8c)$$

With X the biomass concentration, V the liquid volume, μ the specific growth rate, F the feed rate into the reactor, $D = F/V$ the dilution rate, μ_{max} the maximum specific growth rate, and X_{max} the maximum biomass concentration [g/L]. Synthetic experiments were simulated dynamically using a Runge-Kutta $4^{th}/5^{th}$ order ODE solver with $\mu_{max} = 0.3$ h^{-1} and $X_{max} = 47.3$ g/L. Data points were sampled with 1-hour intervals and 5% Gaussian noise (5% of true value). A central composite design (CCD) with 3 factors (initial biomass concentration varying between 0.5-2.5 g/L, initial volume varying between 1.9-3.5 L, and feed rate F varying between 0.05-0.5 L/h) resulted in a data set with 15 experiments (Supplementary File 2).

Given the underlying simplicity, the main objective is to evaluate whether training with a single fed-batch experiment can predict the 14-remaining fed-batch experiments. As such, the data partition consisted of a training subset with a single reactor experiment (the DoE center point, i.e., the 15th experiment corresponding to 24 observations), cross-validation with a single reactor experiment (a repetition of the center point experiment corresponding to 24 observations) and testing with all remaining 14 reactor experiments corresponding to 336 observations.

3.2. Case study 2 – Park & Ramirez fed-batch bioreactor

Case study 2 is a highly nonlinear fed-batch bioreactor of yeast cells expressing a foreign protein with five state variables Park and Ramirez, (1988). The bioreactor is governed by the following set of material balance equations:

$$dX/dt = \mu \cdot X - D \cdot X \quad \text{(9a),}$$

$$dS/dt = -Y \cdot \mu \cdot X + D \cdot (S_{in} - S) \quad \text{(9b),}$$

$$dPm/dt = \theta \cdot (Pt - Pm) - D \cdot Pm \quad \text{(9c),}$$

$$dPt/dt = f_p \cdot X - D \cdot Pt \quad \text{(9d),}$$

$dV/dt = F$ (9e)

The state variables are the biomass concentration (X), glucose concentration (S), secreted protein concentration (PM), total protein concentration (PT), and liquid volume in the reactor (V). This process has a single feed stream with flow rate, F, and substrate concentration $S_{in} = 20\ g/L$. The specific reaction rates are defined by highly nonlinear kinetic equations:

$$\mu = \frac{21.87\, S}{(S + 0.4)(S + 62.5)}$$

(10a),

$$\theta = \frac{4.75\mu}{0.12 + \mu} \quad (10b),$$

$$f_p = \frac{Se^{-5.0S}}{0.1 + S} \quad (10c)$$

With μ the specific growth rate, Y the substrate/biomass yield coefficient, θ the foreign protein secretion rate, and f_p the foreign protein synthesis rate. A total of 16 experiments were simulated across a 0-15 h time window, always with the same initial condition (X(0)= 1.0 g/L, S(0)= 5.0 g/L, PT(0)=0 g/L, PM(0)=0 g/L, V(0)=1.0 L) but with varying feed rate profiles. A Runge-Kutta 4th/5th order ODE solver was adopted for this purpose. Fifteen experiments were generated by a CCD with 3 factors to design step changes in the feed rate profile between 0-2 L/h, namely the feed rate between 0-5 h, F1, feed rate between 5-10 h, F2, and feed rate between 10-15 h, F3. The 16th experiment was simulated using the optimal feed rate profile proposed by Park and Ramirez, (1988) (Supplementary File 3). This experiment resulted in the maximum possible secreted protein production in a 10 L bioreactor (32.4 g) and was used as a test experiment (extrapolation batch).

4. Results

4.1. Development of a bioreactor PINN model for case study 1

The proposed PINN methodology was thoroughly investigated for both bioreactor case studies, starting with the simpler logistic fed-batch reactor problem. The key objective was to evaluate the PINN approach under high data sparsity (training with data from a single experiment (34 observations) and testing with data from the remaining 14 experiments (336 observations)). The data partition was kept the same in all tests performed. The prior knowledge used to compute the physics loss included the biomass and volume material balances Eqs. (8a, b) but excluded the logistic growth Eq. (8c). The data loss included biomass but excluded volume measurements. It was intended to investigate if the PINN can predict unmeasured volume dynamics given that the physics loss Eq. (8b) precisely defines volume dynamics.

To optimize the PINN structure, a systematic hyperparameter search was conducted using Optuna Akiba et al. (2019), an automated framework employing the Tree-structured Parzen Estimator (TPE) algorithm. The optimization targeted key hyperparameters, including the learning rate, activation functions in both networks (tanh, SiLU, and gelu), number of layers (n_layers), nodes (n_units) in FFNN-S, and nodes in FFNN-R (model2_nodes; a single hidden layer was kept given the simplicity of unknown kinetic term defined by Eq. (8c)) over 25 trials. At this stage, the pre-optimized weighting parameter $l = 0.5$ and a number of collocation points, K, equivalent to the size of FFNN-S (K=NWS) (further discussed below) were excluded from the hyperparameter search. Figure (2) depicts a parallel coordinate plot of Optuna's search, where lines represent trials and color indicates objective value. Optuna converged to an optimal PINN configuration with FFNN-S sized as 3x12x12x12x2 with SiLU activation in the hidden layers, FFNN- R sized as 1x3x1 with tanh activation in the hidden layer, and a learning rate of 0.005. This configuration achieved a test error of 0.007.

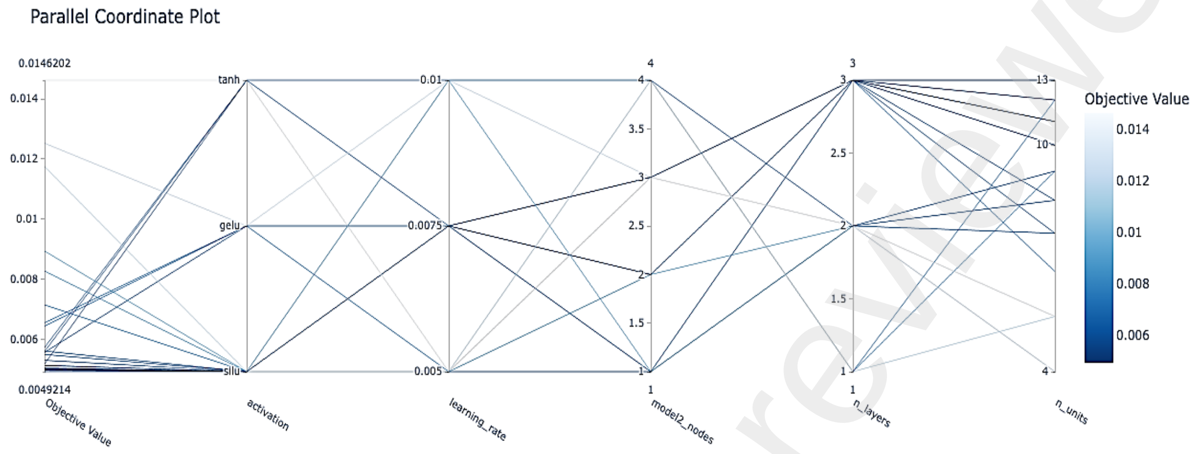


Figure 2: Optuna hyperparameter search results: Parallel coordinate plot showing the relationship between hyperparameters and objective function values for case study 1 over 25 trials.

This automated search was further refined by a manual grid search of key hyperparameters. The SiLU and tanh activation functions derived from Optuna were kept in the manual search. The sizes and learning rate were further investigated. Detailed performance metrics for each PINN variation are presented in (Supplementary File 1, Table S1). The manual refinement converged to an optimal PINN structure (FFNN-S: 3x8x8x8x2 with SiLU, FFNN- R: 1x5x1 with tanh) and a learning rate of 0.0075. The slightly higher learning rate was shown to accelerate convergence without compromising performance. This configuration achieved the same test error of 0.007 of Optuna with a lower number of weights and a lower AICc value.

Figures (3A–D) illustrates the predictive capability and adherence to physics constraints of the selected optimal PINN structure for case study 1. Figures (3A) and

(3B) show the model's performance on the training fed-batch experiment and the 14-test fed-batch experiments, respectively. The coefficient of determination for both biomass (X) and volume (V) is high and comparable for the train and test data subsets, showing excellent predictive power. Moreover, the final train data loss was 0.0046, practically coincident with the noise level (0.0049), showing negligible overfitting. The final test data loss was 0.013, almost twofold the noise level (0.007), but even so very low (Table S1).

Figure (3C) depicts the model's adherence to the underlying physical laws at the randomly generated collocation points. These collocation points were randomly sampled within the design space (biomass, volume, and feed rate) to enforce the physical constraints across the relevant operational range. At each training epoch, a set of collocation points was randomly generated within the known bounds of state variable, ensuring a comprehensive coverage of the design space. The FFNN-S output derivatives (dX/dt and dV/dt) are computed at the collocation points by automatic differentiation and then compared against the derivative values derived from the governing differential equations Eqs. (8a, b). The strong linear correlation observed in Figure (3C) highlights the model's ability to accurately satisfy the governing differential equations. In addition, Figure (3D) extends the analysis of physics against the calculated physics at the test data points. This plot provides critical insight into the model's ability to generalize the physical relationships learned to different regions of the design space. The strong linear correlation observed in Figure (3D) confirms that the model accurately captures the underlying process dynamics across different process regimes.

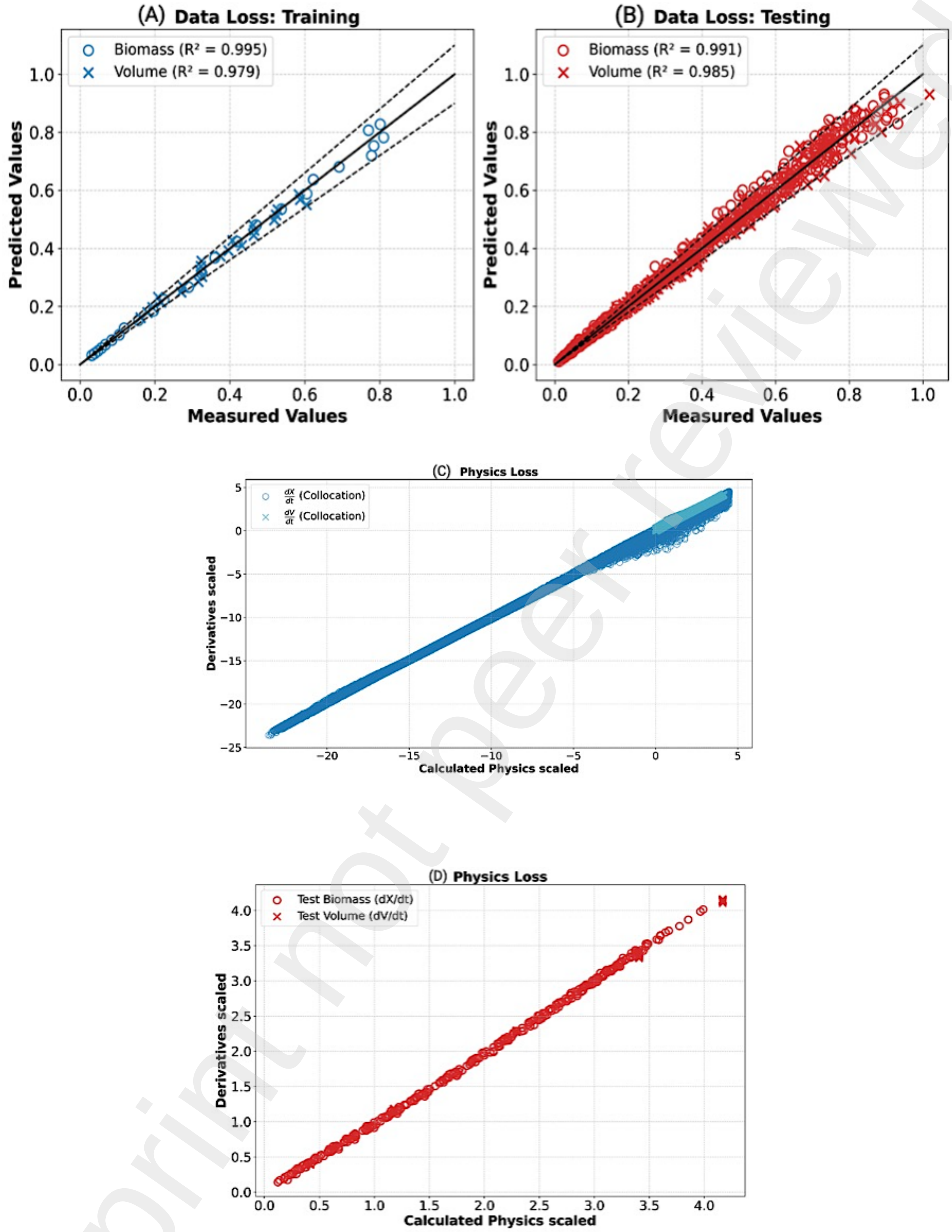


Figure 3: Training results for case study 1 using a PINN structure (FFNN-S: 3x8x8x8x2 with SiLU, FFNN- R: 1x5x1 with tanh) trained with a learning rate of 0.0075, $l = 0.5$ and K= 194 collocation points. **(A)** Predicted versus measured biomass and volume data for the training set, **(B)** Predicted versus measured biomass and

volume data for the testing set, **(C)** Biomass and volume physics at the training collocation points, **(D)** Biomass and volume physics at the test data points.

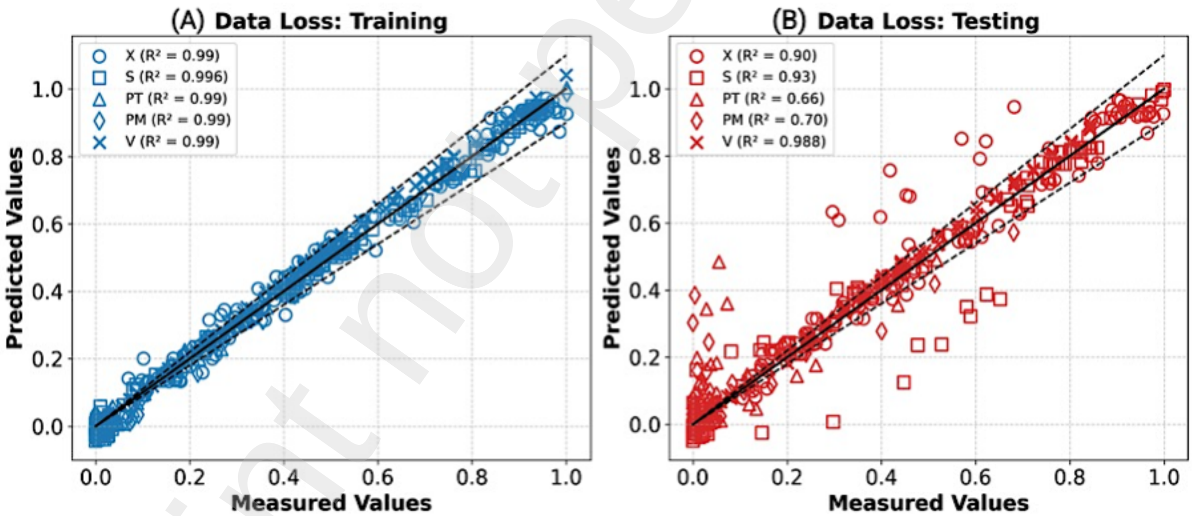
Overall, these results show that the PINN model succeeded in predicting the biomass and volume dynamics of 14 test experiments based on the training data of a single experiment. Notably, the prediction of volume dynamics relied solely on the embedded physics law Eq. (8b). This simple example demonstrates that when full and precise physics of a subset of state variables are known, observations of those state variables are in principle not needed to train the PINN. This highlights a key advantage of PINNs over conventional neural networks since the latter always need data for their training.

4.2. Development of bioreactor PINN model for case study 2

The PINN framework was applied to the more complex Park and Ramirez fed-batch bioreactor benchmark (case study 2), which has a higher dimensionality of state variables (biomass (X), substrate (S), total protein (PT), secreted protein (PM), and volume (V)), highly nonlinear kinetics and time-varying control inputs. The key objective was to evaluate if a PINN trained on a sparse data set (9 of the 15 experiments of the CCD) can predict the optimal production scenario published by Park & Ramirez (32.4 g of secreted protein when the optimal feed profile is applied). The physics loss included the material balance equations of the 5 state variables Eqs. (9 a-e) and excluded the reaction kinetics Eqs. (10 a-c). The same collocation points scheme of case study 1 was employed. The data loss included measurements of all state variables except volume, i.e. X, S, PT, and PM (the volume training relied exclusively on its defining Eq. (9e) as in case study 1). The data loss was based on 9 experiments (the center and square points of the CCD; the star experiments were excluded). Overfitting was prevented by stochastic regularization whereby data from 6 out of 9 experiments were randomly selected to compute the data loss at each epoch. Building on the optimal PINN structural features already identified in case study 1, a systematic grid search of the optimal FFNN-S and FFNN-R sizes was conducted as detailed in (Supplementary File 1, Table S2). This procedure converged to the optimal structure (FFNN-S: 6x17x17x5 with SiLU, FFNN-R: 1x6x3 with tanh).

Figures (4A-D) illustrate the training and prediction performance of the optimal PINN. Figure (4A) shows a strong agreement between predicted and measured values across all five state variables for the training data set. Notably, the coefficient of determination (R^2) exceeded 0.99 for all state variables showing excellent training performance. However, the model's generalization capability degraded significantly when the PINN was exposed to the optimal Park & Ramirez test experiment (Figure, 4B). The (R^2) values for X (0.90), S (0.93), and V (0.99) remained relatively high, but a significant performance degradation was observed for PT, and PM, with (R^2) values of 0.66, and 0.70, respectively. Figure (4C) illustrates the PINN's adherence to the underlying physical laws at randomly generated collocation points. The close alignment along the diagonal in Figure (4C) suggests a high level of adherence to the physics constraints. However, the generalization of physics equations is severely compromised when the PINN is exposed to the test data points (Figure 4D), particularly those associated with protein dynamics (PT and PM), which exhibit a significant scatter.

The Park and Ramirez problem clearly poses a greater challenge for the PINN's ability to generalize than in case study 1. Given the excellent training performance, the degradation of predictive power could be caused by insufficient information content in the training set, as the star points of the CCD-DoE were excluded. To test this hypothesis, the PINN was retrained with the data of the full CCD-DoE (the center, square, and star points of the CCD; 15 experiments). Stochastic regularization randomly selected 10 out of 15 experiments to compute the data loss at each epoch. The overall results are shown in Figures (4 E-F). Indeed, augmenting the training set significantly improved the testing performance. As before, PINN succeeded in describing the full training data set with (R^2) higher than 0.99. The augmented training set likely provided a more comprehensive data coverage enabling the PINN to more accurately learn the intricate nonlinearities. Despite the tenfold decrease in the test loss WMSE (from 0.133 to 0.0128), some scatter persists with a few predictions laying outside the 5% error bound, suggesting there is still room for improvement (further to this in the comparison with hybrid semiparametric below).



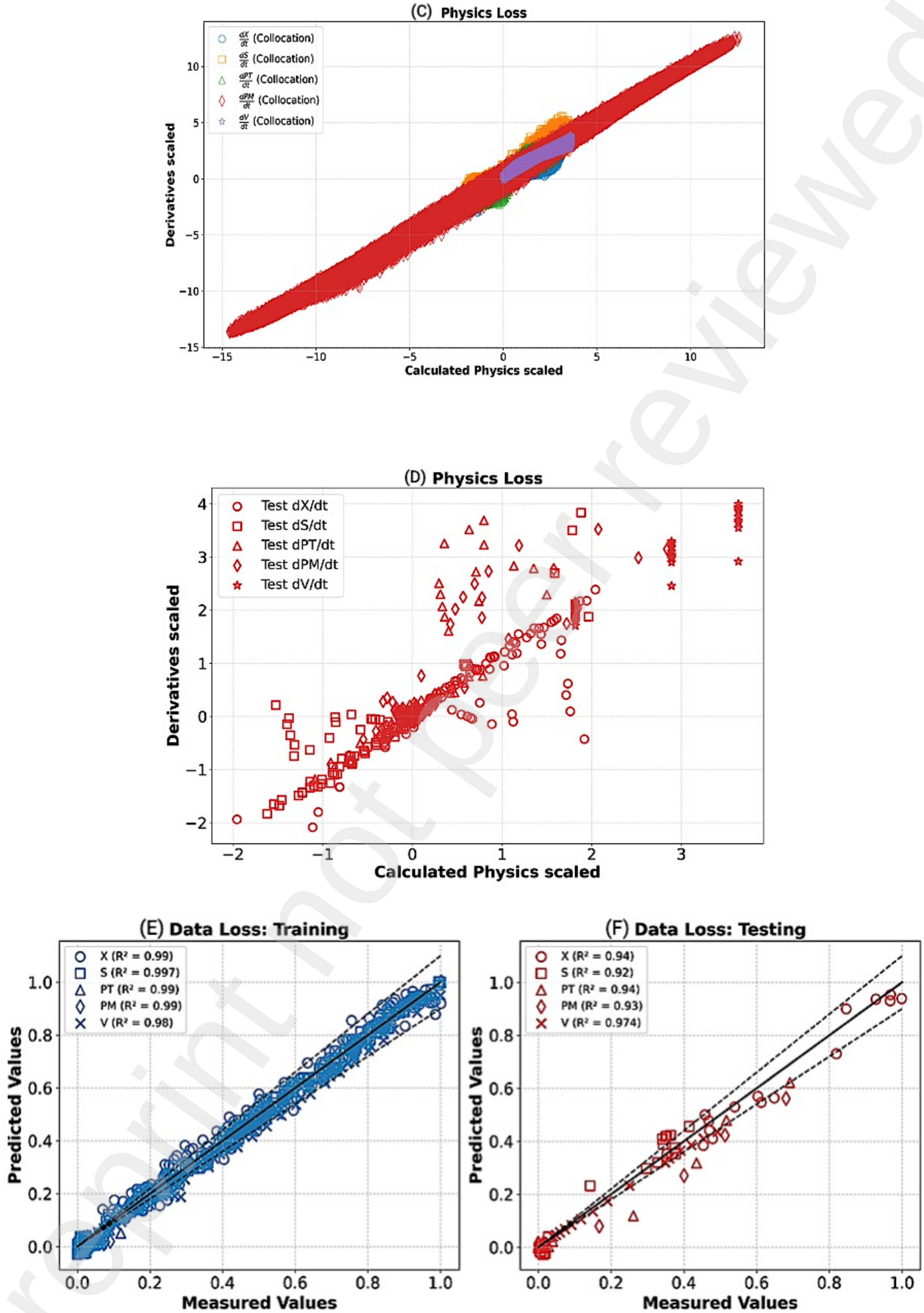
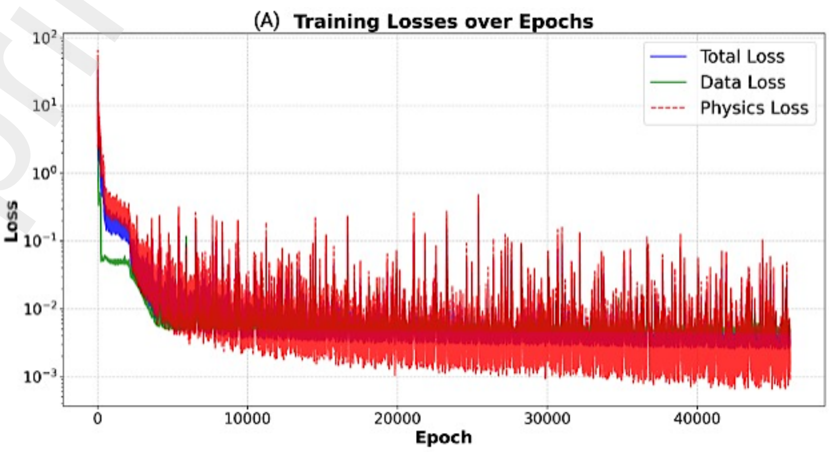


Figure 4: Training results for case study 2 using a PINN structure (FFNN-S: 6x17x17x5 with SiLU, FFNN-R: 1x6x3 with tanh), trained with learning rate 0.0075,

$l = 0.5$ and K= 515 collocation points, when the model is trained over a partial data set with 8 batch experiments (center and square points of the CCD) (A-D) or over the full data set with 15 experiments (full CCD) (E-F). **(A)** Predicted versus measured X, S, PT, PM, V for the partial training set (8 experiments), **(B)** Predicted versus measured X, S, PT, PM, V for the testing set (optimal Park & Ramirez bioreactor with 32.4 g production), **(C)** X, S, PT, PM, V physics at the training collocation points, **(D)** X, S, PT, PM, V physics at the test data points, **(E)** Predicted versus measured X, S, PT, PM, V when the PINN is trained on the full CCD (15 experiments), **(F)** Predicted versus measured X, S, PT, PM, V for the testing set (optimal Park & Ramirez bioreactor with 32.4 g production).

4.3. Training convergence

The training of PINNs is complicated by the need to simultaneously minimize the training and physics loss terms. This study followed the most common approach whereby a total loss function, defined as the weighted sum of physics and data loss terms Eq. (6), is minimized. In this approach, it is critically important that both loss terms are correctly normalized. Figures (5 A-B) elucidates the training convergence behavior of the optimal PINN structures for case studies 1 and 2, which includes the evolution of the data loss, physics loss, and total loss across training epochs.



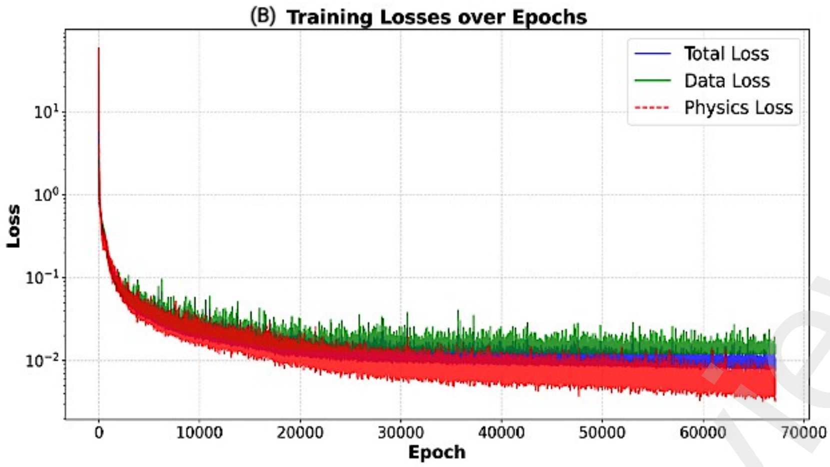


Figure 5: Loss function evolution as function of training epoch with a learning rate of 0.0075, $l = 0.5$ and K/NWS=1. **(A)** PINN structure (FFNN-S: 3x8x8x8x2, FFNN-R: 1x5x1) for case study 1, **(B)** PINN structure (FFNN-S:6x17x17x5, FFNN-R:1x6x3) for case study 2

In both cases a stable although noisy convergence behavior is observed, characterized by a rapid initial decline in the total loss function, encompassing both data and physics losses, followed by a prolonged plateau at low loss values. In case study 1, stochasticity is introduced in the physics loss function via the random selection of collocation points, which propagates to the data loss function via the update of neural network parameters. In case study 2, stochasticity originates from both the physics loss (random collocation points) and data loss (random experiments to compute the data loss). The physics loss term tends to plateau always below the data loss term. This is explained by the random experimental error that is intrinsic to the data residuals but absent in the physics residuals. Importantly, the data loss terms plateaus close to the WMSE noise level (0.0049 for case study 1 and 0.013 for case study 2) showing that in both cases overfitting was successfully mitigated.

4.4. Relative importance of data and physics loss

The weighting parameter λ sets the relative importance of the physics loss (λ) and data loss (1 - λ). The choice of λ may significantly influence the training convergence and overall model performance. The studies above assumed equal importance of physics and data loss terms ($\lambda = 0.5$). Figure (6A) shows the effect of λ on the final train and test losses for the optimal PINN structure for case study 1. At λ = 0.0, the physics loss is not minimized thus the PINN becomes analogous to a conventional neural network model. As expected, with λ = 0.0 the PINN succeeded in describing the training data (low final data loss). However, the test data loss was the highest among all runs performed. This highlights the importance of "physics learning" for the PINN's ability to extrapolate to unseen observations. As λ increases from 0.25 to 0.9, a clear decrease trend in the train and test physics is observed. Notably, this is accompanied by a decreasing trend in the test data loss denoting synergies between physics and data learning. Setting λ = 0.5 resulted in the lowest loss values, thus providing the optimal trade-off between data and physics loss terms. Figure (6B) shows similar results for the optimal PINN structure of case study 2. Likewise, λ = 0.5 provided the optimal balance between data and physics loss minimization. All in all, these results demonstrate the synergy between the data and physics learning components in the PINN framework. Learning physics is shown to be critically important for the PINN's extrapolation capacity, proving a clear advantage over conventional neural models (further discussed below).

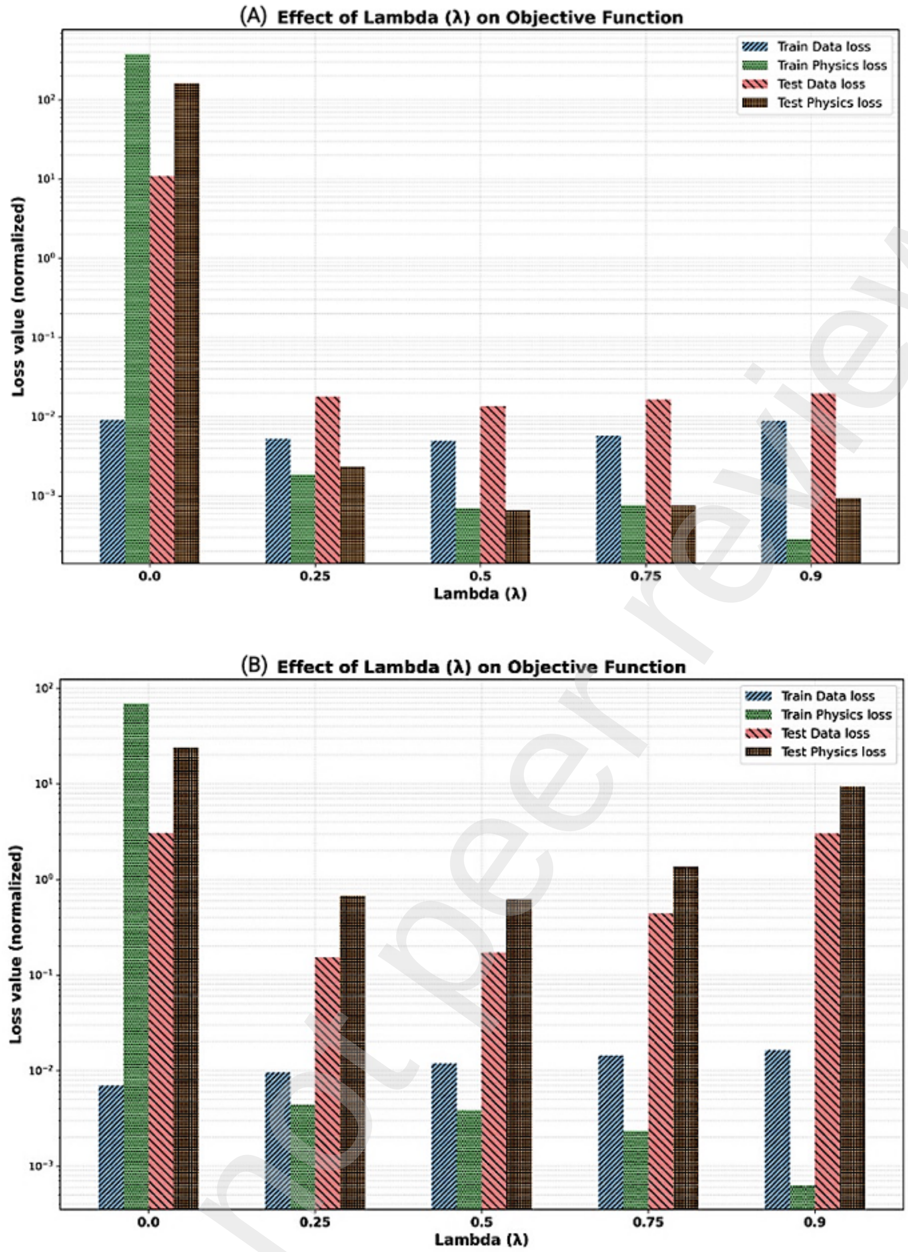
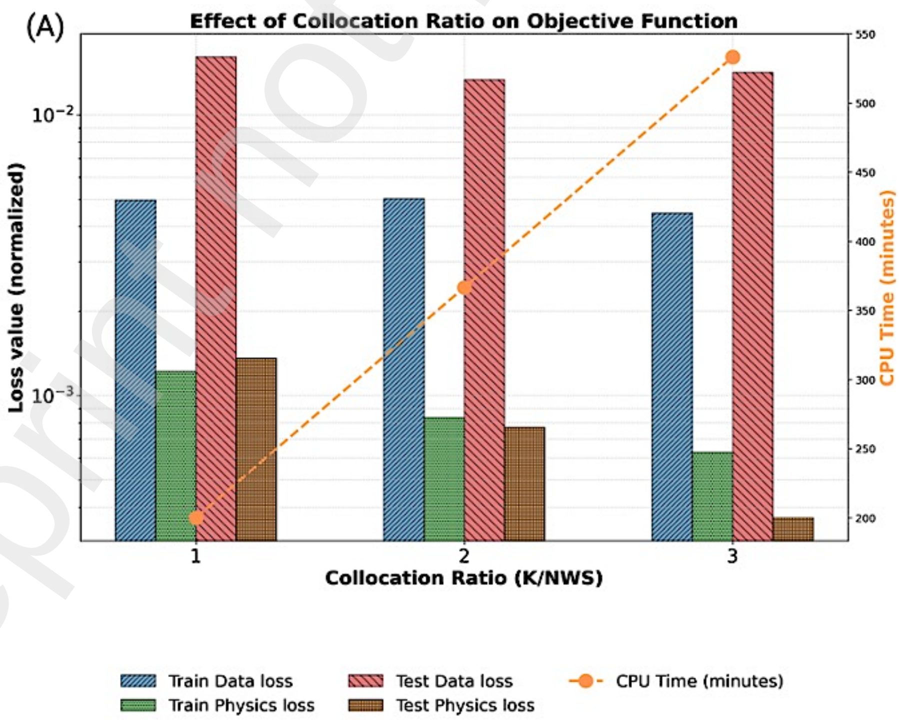


Figure 6: Dependency of data and physics losses for different λ values. **(A)** Optimal PINN structure for case-study 1 (FFNN-S: 3x8x8x8x2, FFNN-R: 1x5x1), **(B)** Optimal PINN structure for case-study 2 (FFNN-S:6x17x17x5, FFNN-R:1x6x3).

4.5. Effect of collocation points

The number of collocation points (K) directly influences the physics loss minimization and indirectly the training convergence and overall model metrics. In this study, the number of collocation points was set as multiples of the number of weights (NWS) of the FFNN-S. The ratio K/NWS = 1 thus provides a sufficient number of physics residuals to train FFNN-S, even when no data points are available. Increasing K/NWS could in theory positively influence the training convergence. Figures (7A-B) shows the effect of increasing the ratio K/NWS = 1, 2, 3 on the train and test loss terms for the optimal PINN structures of case studies 1 and 2. As anticipated, increasing K/NWS generally reduces the train and test physics loss term, with however a more marked improvement in case study 1 than in case study 2. The physics loss reduction is however not reflected in the data loss terms, which seem to be relatively insensitive to the K/NWS ratio. It was concluded that a K/NWS =1 is sufficient for achieving robust model convergence. Importantly, the computational cost (CPU) escalates linearly with increasing K/NWS. Consequently, a ratio of 1 presents an optimal balance between model accuracy and computational efficiency. For this reason, K/NWS=1 was set in all tests performed in the present study.

(A)



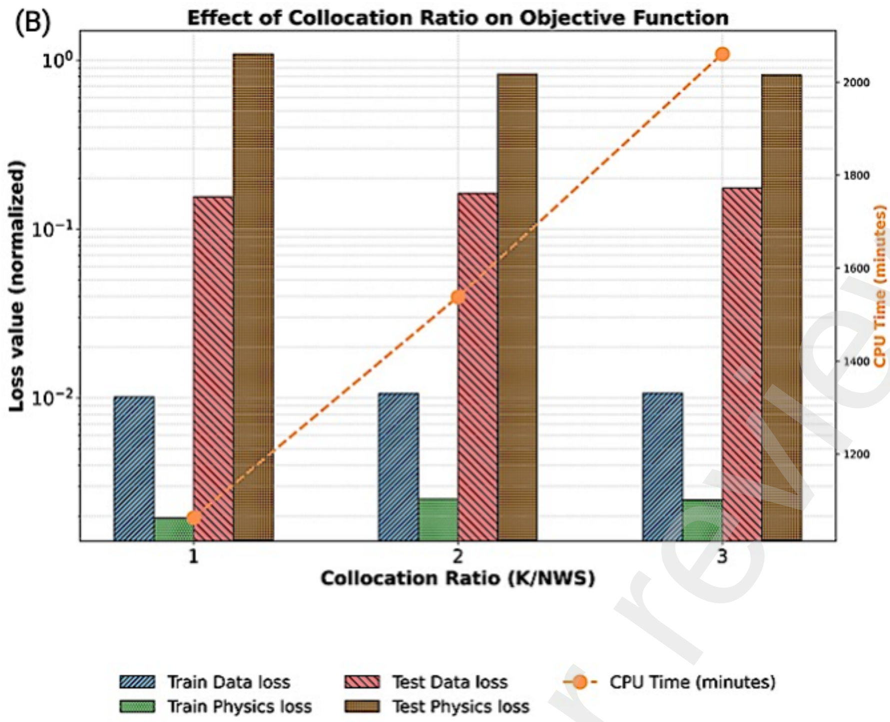


Figure 7: Dependency of data and physics losses for different K/NWS collocation points. **(A)** Optimal PINN structure for case-study 1 (FFNN-S: 3x8x8x8x2, FFNN-R: 1x5x1), **(B)** Optimal PINN structure for case-study 2 (FFNN-S:6x17x17x5, FFNN-R:1x6x3).

4.6. Comparison with hybrid semiparametric models

Several studies have shown the superiority of PINNs in relation to conventional neural networks for bioreactor problems (e.g., Bangi et al., 2022; Moayedi et al., 2024; Velioglu et al., 2025). Only a few studies compared PINNs and hybrid semiparametric modeling (Yang et al., 2024; Jul-Rasmussen et al., 2025). It is particularly interesting to analyze if both approaches perform differently when exposed to the same levels of data and prior knowledge. To assess this, optimal PINN structures were compared to corresponding optimal hybrid semiparametric structures and standard neural network structures (without the physics component) for both case studies. Furthermore, the dual-FFNN PINN concept proposed in this study was compared with an equivalent single-FFNN PINN where all state variables and kinetic rates are computed by a single FFNN. The optimal sizes of the FFNN-S and FFNN-R for all model types were systematically evaluated by manual search as previously performed for the dual-FFNN PINN structure. To ensure a fair comparison, the 4 types of models were subject to identical training, validation, and testing processes. Detailed performance metrics for each type of model are given in (Supplementary File 1, Tables S1 and S2 for case studies 1 and 2 respectively). A summary of the results is presented in Table 1.

Table 1 – Comparison of best Dual-FFNN PINN, Single-FFNN PINN, conventional FFNN and Hybrid semiparametric models for case studies 1 and 2

Model	FFNN-S	FFNN-R	Train Data loss WMSE	Validation WMSE	Test Data loss WMSE	AICc	Nw	CPU (min)
Case study I								
Dual-FFNN PINN	3x8x8x8x2	1x5x1	0.0046	0.0054	0.013	-1895.7	210	109.17
Single-FFNN PINN†	3x8x8x8x3	NA	0.0053	0.0047	0.258	-1856.78	203	104.9
Conventional FFNN‡	3x3x1	NA	0.0046	0.007	168.83	190.83	16	5.62
Hybrid semiparametric *	NA	1x5x1	0.0046	0.0085	0.007	-19.44	16	103.9
Case study II								
Dual-FFNN PINN	6x17x17x5	1x6x3	0.012	NA	0.133	-11766.50	548	732.3
Single-FFNN PINN†	6x17x17x8	NA	0.010	NA	0.177	-14057.06	569	810
Conventional FFNN‡	6x5x5x4	NA	0.011	NA	0.521	-102.83	89	125.2
Hybrid semiparametric*	NA	1x6x3	0.013	NA	0.039	-108.26	33	702

(†) Single-FFNN PINNs consisted of a FFNN-S structure where the output layer computes all the state variables and all the unknown kinetic rates (3 outputs for case study 1 and 8 outputs for case study 2)
(‡) The conventional neural network models consisted of a single FFNN-S structure trained with data of measured state variables without the physics loss
(*) The hybrid semiparametric models followed the general structure of Figure (1B). The material balance ODEs were inserted directly in the model structure in replacement of the FFNN-S, namely

Eqs. (8a, b) for case study 1 and Eqs. (9a-e) for case study 2. The reaction kinetics were considered unknown thus described by the FFNN-R similarly to the dual-FFNN PINN structure

Table 1 shows that the dual-FFNN PINN structure outperforms a single-FFNN structure with a comparable number of parameters in terms of predictive power for both case studies. The test WMSE was 19.4-fold higher in case study 1 and 33% higher in case study 2. The same conclusion can be taken for all structures tested without exception (Supplementary File 1, Tables S1 and S2). In most cases, the train WMSE levels off at higher values in the case of single-FFNN PINNs compared to the dual-FFNN PINNs of the same size. The uncoupling of state parameterization and kinetics parameterization seems to be advantageous for the training stability. In what follows the analysis of results is focused on the dual-FFNN PINN structure.

For case study 1, the dual-FFNN PINN, hybrid semiparametric, and conventional FFNN models achieved the same final training WMSE (Table 1). Of relevance, both the dual-FFNN PINN and semiparametric approaches utilized an equally-sized FFNN-R to approximate the specific growth rate (μ) as a function of biomass concentration (X), with a single hidden layer with 5 tanh nodes. The prediction accuracy of the hybrid semiparametric approach was however superior to that of the PINN (~50% lower test WMSE). Despite the twofold higher test error, the modified AICc criterion favored the PINN over the hybrid semiparametric approach because of the high number of physics residuals (parameter K in Eq. (7)). Thus, the AICc criterion appears to be unsuitable to fairly discriminate PINN models. The test WMSE of the conventional FFNN model is several orders of magnitude higher than the other approaches.

Figures (8A-B) further detail the prediction profiles for an exemplary test experiment (the one with the highest feed rate). The time scale was extended to 166 hours to further evaluate the temporal extrapolation capabilities of the models. Both the dual-

FFNN PINN and semiparametric models effectively approximated the growth curve within the scatter of the experimental data. In contrast, the conventional FFNN severely failed to predict the true dynamic trajectory. When evaluated under extended temporal extrapolation conditions, the hybrid semiparametric model converges to a stable final biomass concentration, although with an offset that in this case amounted to 8% underprediction of the true value. The PINN exhibits, however, unstable behaviour as it diverges from the final biomass concentration over time. This apparent disadvantage of the PINN is not circumstantial as the same behaviour was replicated in several other experiments. Similar conclusions can be reached in the case of volume (V) Figure (8B). The hybrid semiparametric model shows a perfect volume prediction because the full physics is embedded in the model, thus the only source of error stems from the numerical integration. Conversely, the PINN shows an increasing error over time most likely due to limitations in the extrapolation of the learned $dV/dt=F$ equation. Of note, the ANN does not predict volume due to the unavailability of volume data points to train it. Figures (8C-D) show the extrapolation WMSE for all test batches over the extended timeframe confirming that the hybrid semiparametric model is far superior to the PINN. It is worth noting that training with a single experiment configures an extreme data sparsity scenario. Increasing the number of training batches to two significantly reduced the hybrid semiparametric offset and significantly improved the PINN extrapolation performance (results not shown).

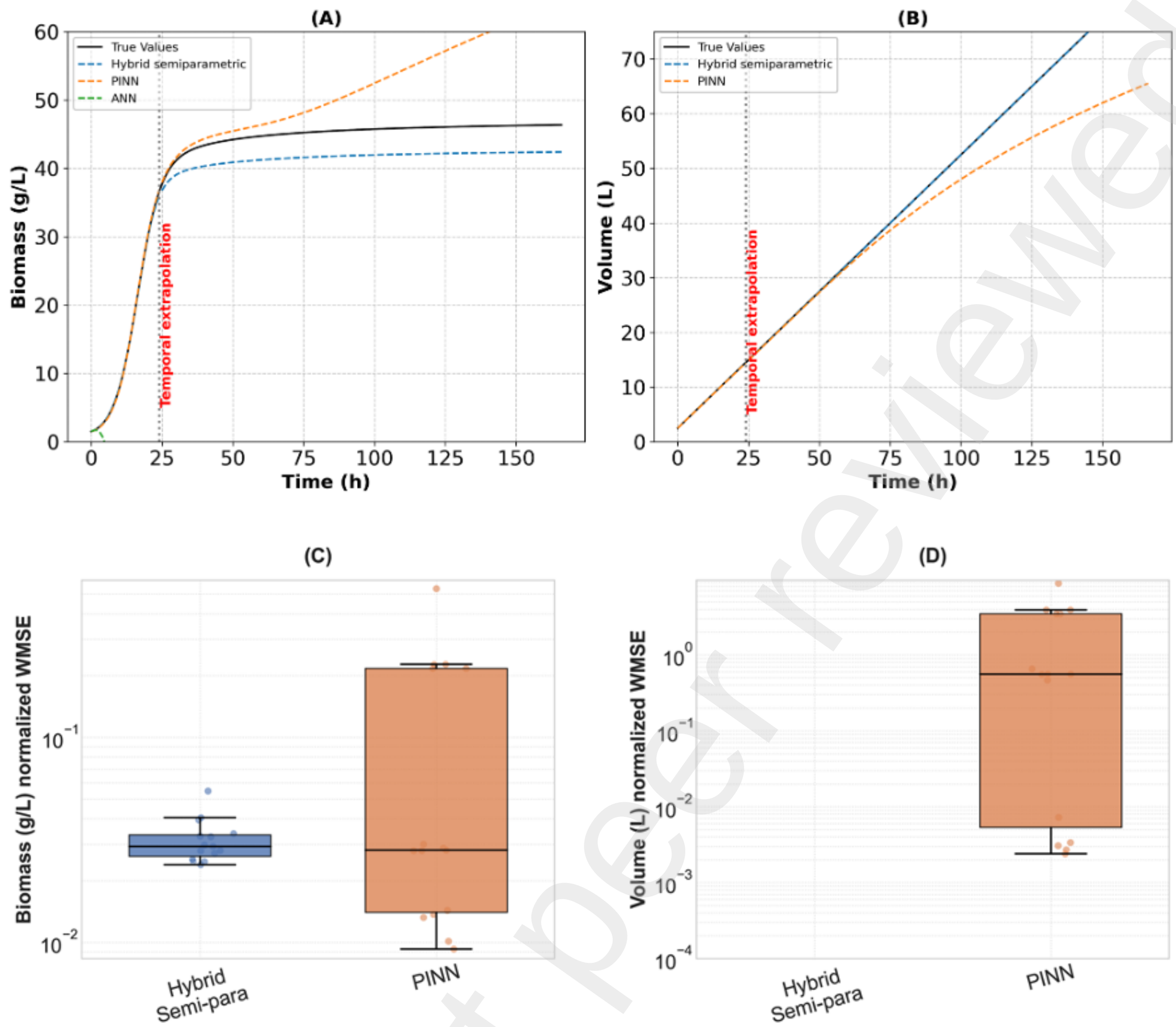


Figure 8: Compares the dual-FFNN PINN (FFNN-S: 3x8x8x8x2, FFNN-R: 1x5x1) and hybrid semiparametric (ODE(2); FFNN-R: 1x5x) extrapolation capacity for case study 1. **(A)** Biomass prediction for the test experiment 10 (highest feed rate) over an extended timeframe 0-166 h, **(B)** Volume prediction for the test experiment 10 (highest feed rate) over an extended timeframe 0-166 h, **(C)** Biomass WMSE distribution across all (training + testing) batches for the extended temporal extrapolation (0 - 166 h), **(D)** Volume WMSE distribution across all (training + testing) batches for the extended temporal extrapolation (0 - 166 h).

A similar analysis was performed for case study 2. The FFNN-R structure of the hybrid semiparametric model and the FFNN-S structure of the conventional FFNN model were systematically evaluated by grid search (Supplementary File 1, Table S2). A summary of results for the best structures is given in Table 1. As for case study 1, the PINN, hybrid semiparametric, and conventional FFNN models achieved practically the same final training WMSE, denoting a comparable descriptive power of the training data. The prediction accuracy of the Park & Ramirez optimal experiment is however very divergent among the 3 models. The hybrid semiparametric model achieved the lowest test WMSE of 0.039, followed by the PINN with a 3.4-fold increase. As expected, the conventional FFNN model had the lowest prediction accuracy, showing a 13.1-fold increase in the test error in relation to the hybrid semiparametric model.

Figures (9A-F) depict the predicted profiles of biomass (X), substrate (S), total protein (PT), secreted protein (PM), and volume (V) for the optimal Park & Ramirez feed rate profile. Figure (9F) shows the piecewise constant optimal feed rate profile (1h step size) that delivers optimal protein production ($PM(15) \times V(15) = 32.4g$ at 15h cultivation time). The hybrid semiparametric model provides a significantly better fit to the true process dynamics. The final prediction of secreted protein production was 32.48 g. In contrast, the PINN shows significant underperformance compared to the hybrid semiparametric model, particularly in the predictions for (S, PT, PM, V). The final prediction of secreted protein was only 28.16 g. The conventional FFNN showed the highest deviations from the true profiles. As for case study 1, the FFNN model was not trained with volume data thus it is unable to predict volume. Importantly, the FFNN model occasionally predicted negative concentration values of S, PT, and PM. This problem was largely mitigated in the case of PINN due to the physics regularization during the training except for substrate. Substrate poses a significant challenge because it depletes to almost zero concentration in the last 4 cultivation hours. The PINN could not cope with negative concentrations most likely to inaccurate material balance equations extrapolation (inaccuracies in the learned physics). The hybrid semiparametric model practically eliminated the negative concentrations as the material balance equations are numerically integrated.

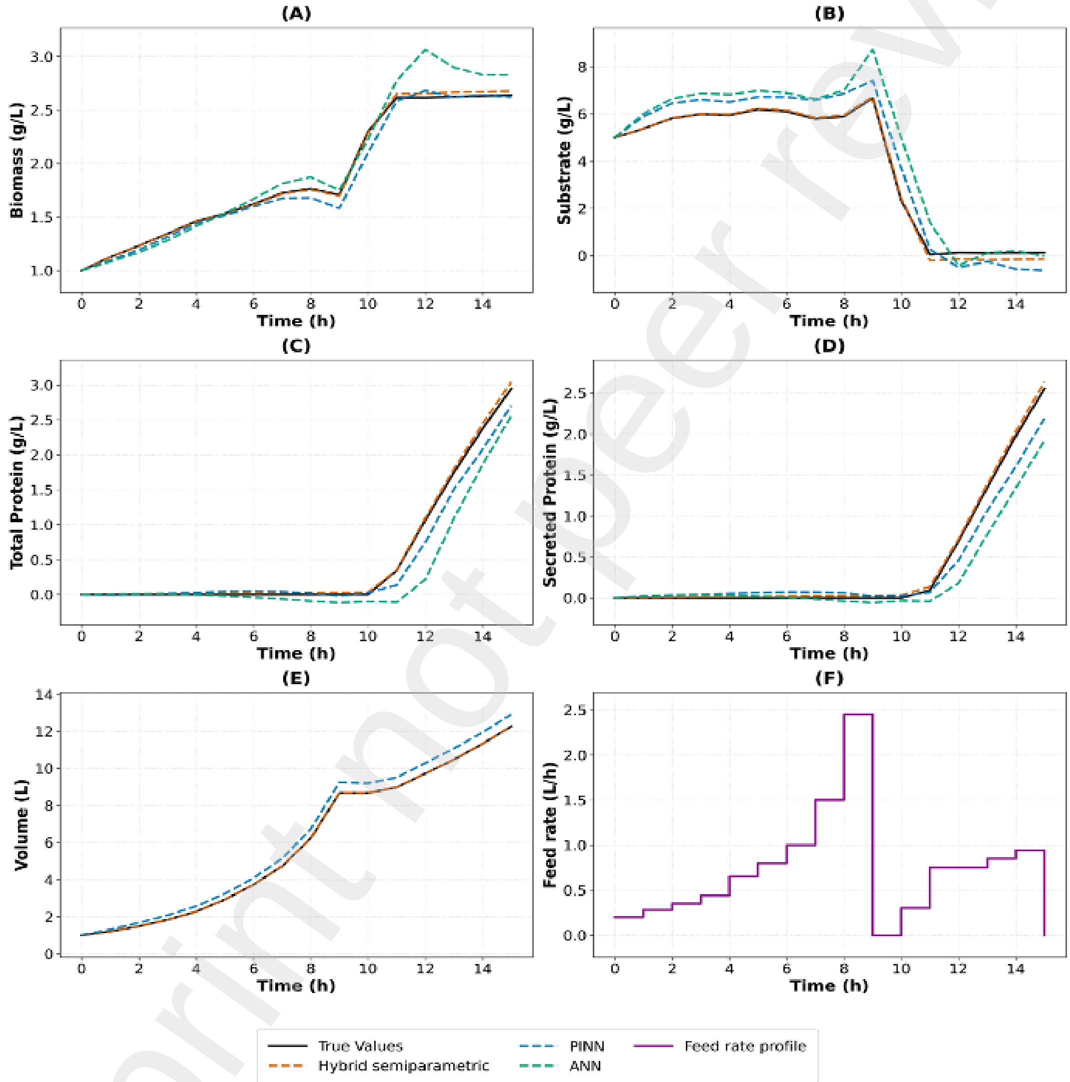


Figure (9 A - F): Compares the best dual-FFNN PINN, best hybrid semiparametric, and best FFNN models (Table 1) for the Park & Ramirez optimal production experiment (test data set). **(A)** Biomass overtime, **(B)** Substrate overtime, **(C)** Total protein overtime, **(D)** Secreted protein overtime, **(E)** Volume overtime, **(F)** Optimal feed rate profile.

5. Discussion

This study presents a comparison of PINNs and hybrid semiparametric modelling for bioprocesses, evaluated across two case studies with varying levels of mechanistic complexity. The comparative analysis elucidates the strengths and limitations of each modelling paradigm, particularly under data-scarce regimes and extrapolative tasks. In case study 1, which involved logistic microbial growth governed by relatively simple dynamics, both the PINN and hybrid semiparametric models achieved excellent predictive performance within the data domain. The PINN effectively inferred unobserved volume dynamics using only biomass measurements, showcasing the potential of embedding physical knowledge within the loss function to predict unobserved target variables. However, as shown in Figure (8), the hybrid semiparametric model consistently outperformed the PINN in extended temporal extrapolation, benefiting from its direct integration of governing equations within the model structure.

Case study 2 posed a significantly more complex scenario, involving five interacting state variables, nonlinear reaction kinetics, and time-varying control inputs. Under these conditions, the performance of the PINN deteriorated notably, particularly in extrapolating the dynamics of secreted protein (PM) and total protein (PT) for sparse data sets. The hybrid semiparametric model again outperformed the PINN model, offering lower train and test WMSE values and superior fit across all variables. The PINN struggled to generalize effectively to unseen feed profiles, a result that is consistent with literature studies that reported PINN shortcomings in capturing nonlinear dynamics due to gradient pathologies and the stiffness in the optimization process, where different parts of the solution space vary on different scales (Wang et al., (2021); Wang et al., (2022)). Li and Feng (2022) emphasized the importance of balancing physics and data losses during training, a challenge that our study addressed through manual tuning of the weighting parameter λ as discussed earlier in (Section 4.4). By contrast, the hybrid semiparametric model maintained stable and accurate performance across both case studies. Its architecture, that incorporates differential equations directly in the model structure, enabled transparent integration of mechanistic knowledge, reduced training complexity, and improved convergence. Similar results were reported by Jul-Rasmussen et al. (2025).

Unsurprisingly, the conventional FFNN consistently underperformed in both case studies due to its lack of mechanistic structure, displaying high variance and poor generalization comparatively to the PINN and hybrid semiparametric models. These findings are supported by the recent work of Velioglu et al. (2025)**,** who proposed a differential-algebraic equation based heuristic to determine whether a PINN can estimate hidden states in partially known systems. Their results demonstrated the superior extrapolation and state reconstruction capabilities of PINNs compared to conventional neural networks, even in the absence of full constitutive equations. This aligns with our extrapolation study where the PINN has accurately extrapolated beyond the training domain even for unobserved states such as the case of unmeasured volume. However, this extrapolative advantage is not guaranteed in more complex systems as highlighted in case study 2.

Recent methodological contributions further insight into PINN limitations. Subramanian et al. (2023) proposed adaptive collocation schemes that target high-error regions to improve physics learning and physics extrapolation. While our study did not adopt adaptive resampling, we employed a stochastic collocation strategy that regenerated collocation points at each training epoch across the bounded state space. This approach proved effective for enforcing broad physical coverage and stabilizing the physics loss. To avoid overfitting, Gaussian Process (GP) smoothening strategies were suggested by Bajaj et al. (2023). This study followed conventional approaches, for instance, case study 1 employed cross-validation using repeated center-point experiments, while case study 2 leveraged stochastic regularization via minibatch size data resampling. Furthermore, our reliance on automatic differentiation (AD) aligns with conventional PINN formulations, but results from Chiu et al. (2022) suggest that hybrid differentiation strategies by coupled automatic-numerical differentiation method (Can-PINN) enhance computational efficiency and accuracy in solving complex physics systems, an avenue worth exploring in future work.

Overall, this study highlights the nuanced capabilities of each modelling framework. Hybrid semiparametric models offer high accuracy and stability within well-defined process regimes. This is particularly true when the backbone of the model is precisely defined by a mechanistic equation, such as Eqs. (1a,b) for bioreactor problems. A key advantage is that the extrapolation of physics is not endangered by training limitations. The requirement of numerical integration may, however, constitute a limitation, particularly for stiff ODE systems. PINNs also have high potential to learn from data and physics simultaneously and may be more flexible for problems with partially known physics. They show a clear advantage over conventional ANNs, but their multi-objective training is far more complex. PINNs do not suffer from the ODEs stiffness limitations of hybrid semiparametric models. Their critical disadvantage respects the extrapolation of physics, which is not guaranteed for the case of PINNs. The choice between these paradigms should thus be dictated by the modelling objective whether high-fidelity interpolation or robust extrapolation is required, by the nature of available process knowledge and encountered numeric implementation hurdles.

6. Conclusion

This study compared PINN, hybrid semiparametric, and conventional ANN modelling for generic bioreactor modelling illustrated with 2 case studies. The results highlight that PINNs, through their incorporation of physics in the loss function, exhibit stronger extrapolation capabilities than conventional ANNs, particularly in scenarios of high data sparsity. Hybrid semiparametric models, which embed the physics equations directly in the model structure, tend to deliver superior predictive accuracy and convergence stability than PINNs for highly nonlinear dynamical systems. The findings emphasize that while PINNs provide a flexible framework for integrating partial prior knowledge, their performance is sensitive to system complexity, hyperparameter tuning, and training instabilities. Hybrid models, in contrast, offer interpretability and robustness but may be less adaptable in extrapolative scenarios where physics may also change. ANN models, lacking any form of physical constraint, consistently underperform in both predictive accuracy and generalization. This study underscores the importance of a problem-driven modelling strategy. PINNs are well-suited for model development under data limitations and structural uncertainty, while hybrid semiparametric models are preferable for process domains with broader mechanistic insights without numerical hurdles.

Abbreviations

FFNN-S, State variables neural network; FFNN-R, Reaction kinetics neural network; NW, Number of network weight; K, Number of collocation points; SiLU, Sigmoid weighted linear unit; tanh, hyperbolic tangent; ADAM, Adaptive Moment Estimation; WMSE, Weighted mean squared error

CRediT authorship contribution statement

Monesh kumar Thirugnanasambandam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **José Pinto:** Writing – review & editing, Methodology, Investigation, Supervision, Conceptualization. **Ekaterina Moskovkina:** Writing – review & editing, Investigation. **Rafael S. Costa:** Writing – review & editing, Methodology, Conceptualization, Supervision. **Rui Oliveira:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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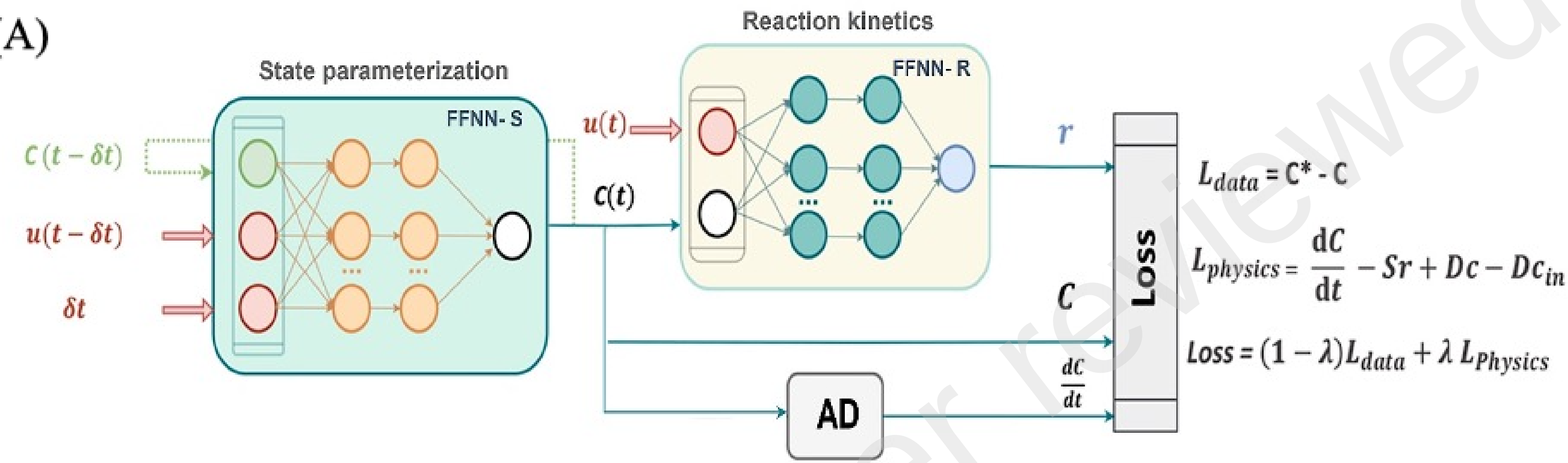
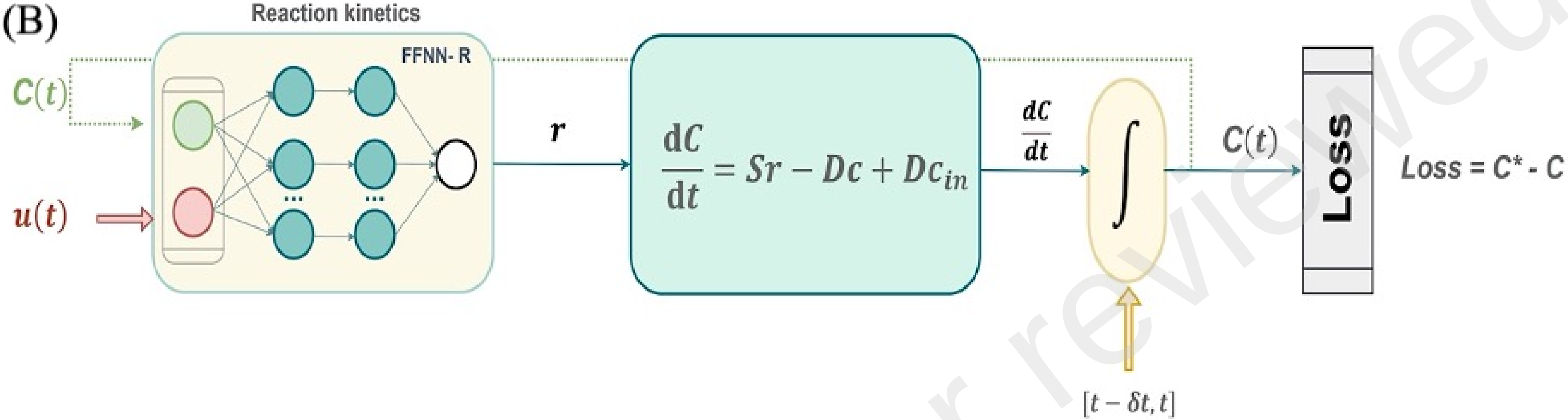
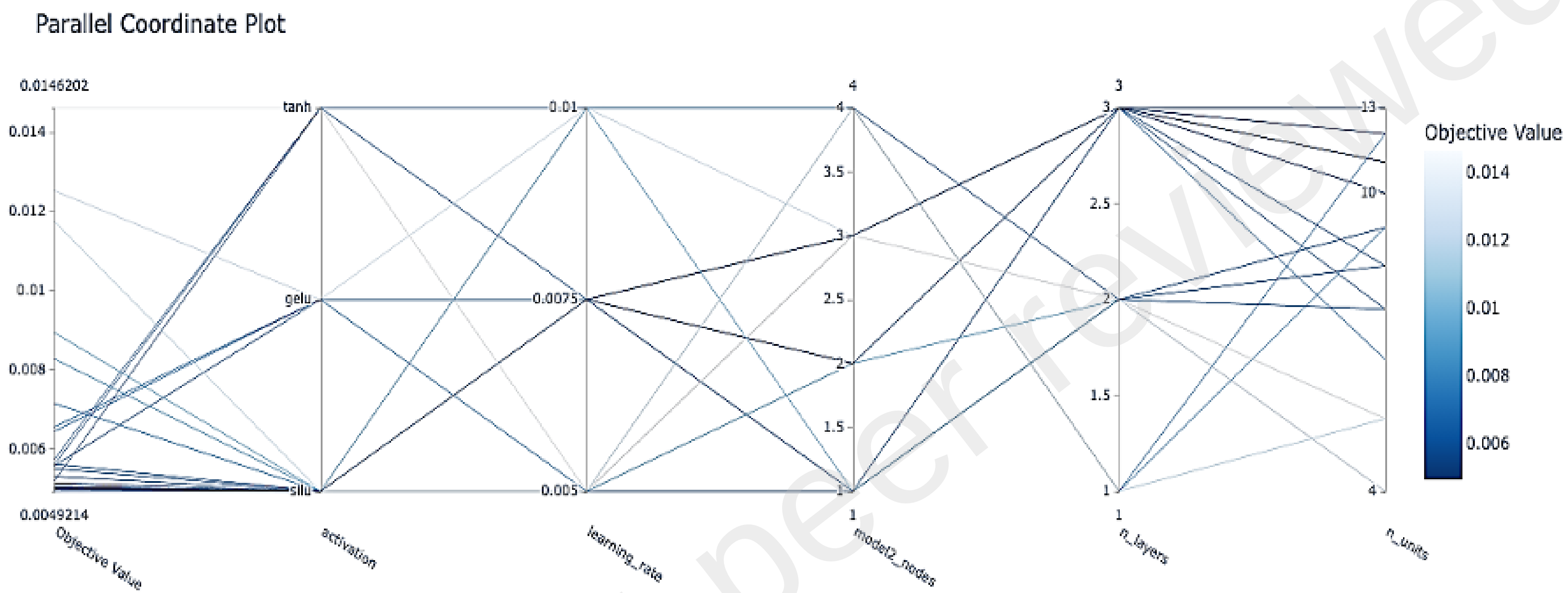
(A)

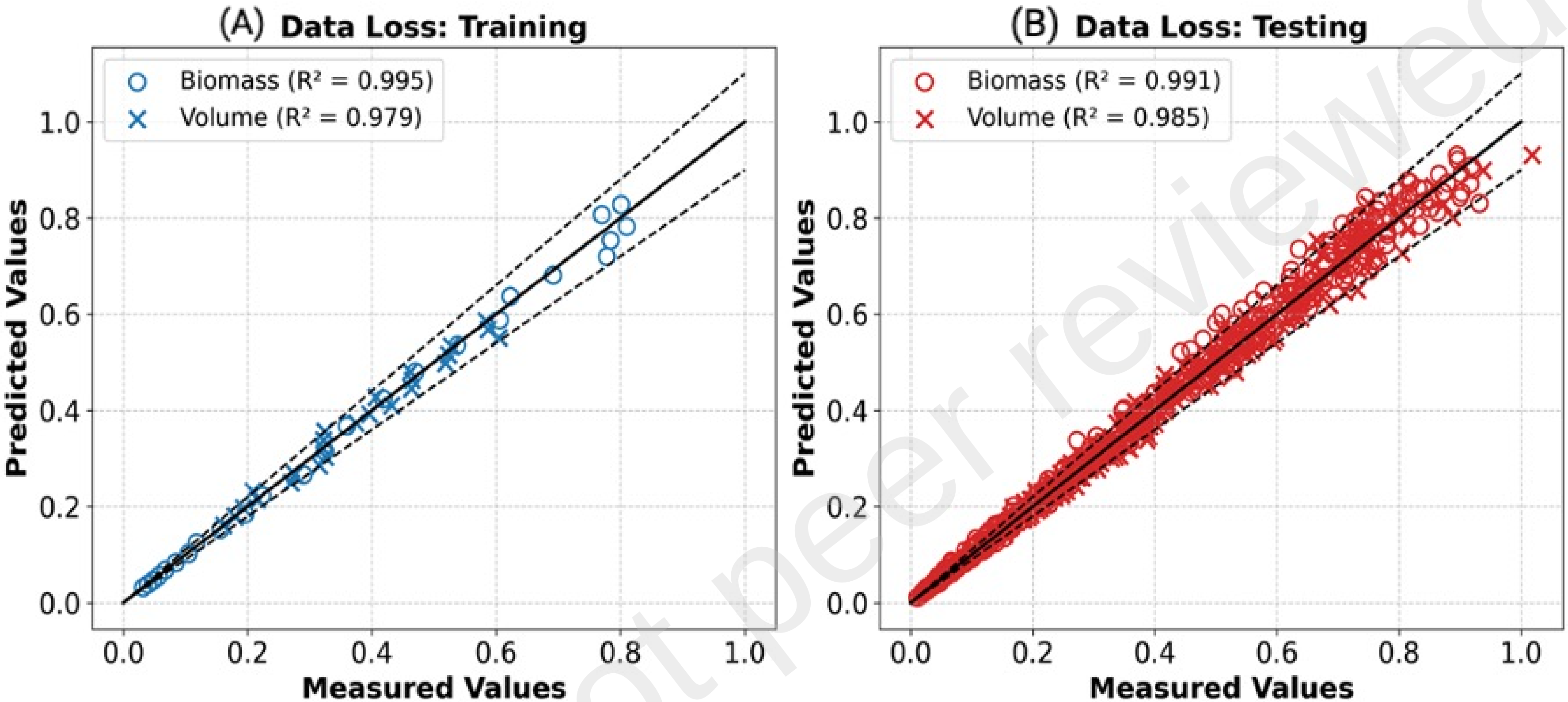

Reaction kinetics
State parameterization
FFNN- S
FFNN- R
$c(t-\delta t)$
$u(t-\delta t)$
δt
$u(t)$
$c(t)$
r
C
$\frac{dC}{dt}$
AD
Loss
L_{data} = C* - C
$L_{physics} = \frac{dC}{dt} - Sr + Dc - Dc_{in}$
$Loss = (1-\lambda)L_{data} + \lambda\, L_{Physics}$

(B)

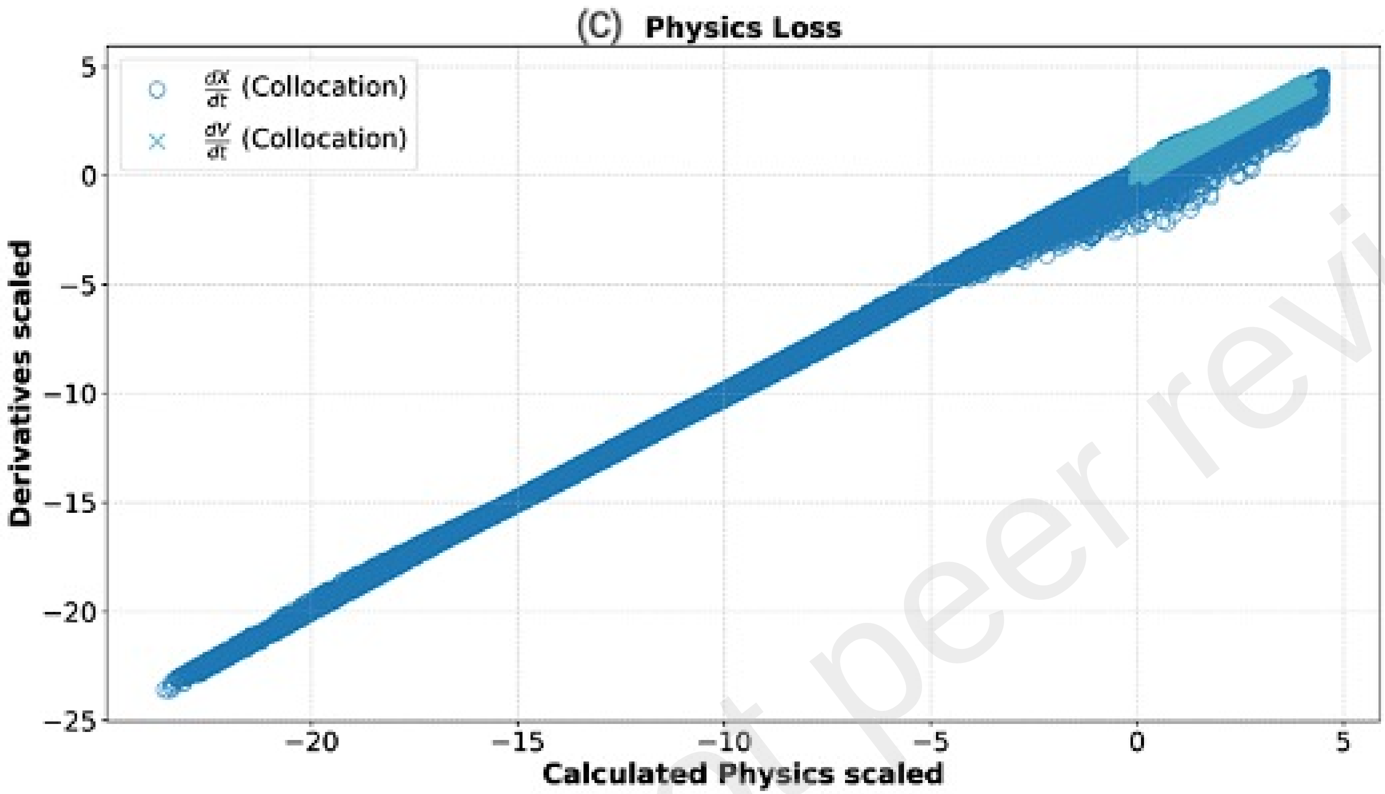


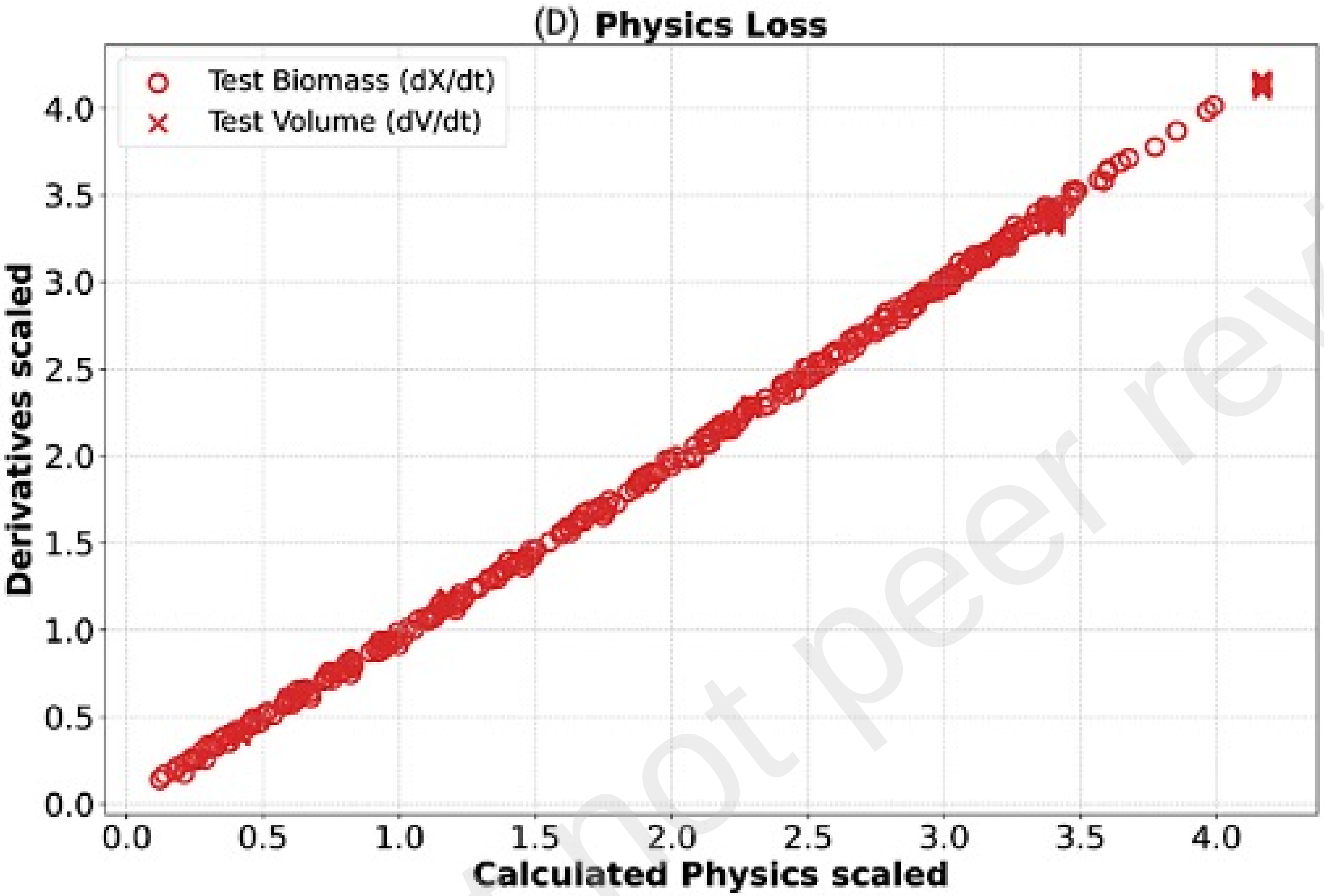
Reaction kinetics
FFNN- R
$C(t)$
$u(t)$
r
$\frac{dC}{dt} = Sr - Dc + Dc_{in}$
$\frac{dC}{dt}$
$C(t)$
Loss
Loss = C* - C
$[t - \delta t, t]$


Parallel Coordinate Plot
0.0146202
0.014
0.012
0.01
0.008
0.006
0.0049214
Objective Value
tanh
gelu
silu
activation
0.01
0.0075
0.005
learning_rate
4
3.5
3
2.5
2
1.5
1
model2_nodes
3
2.5
2
1.5
1
n_layers
13
10
4
n_units
Objective Value
0.014
0.012
0.01
0.008
0.006

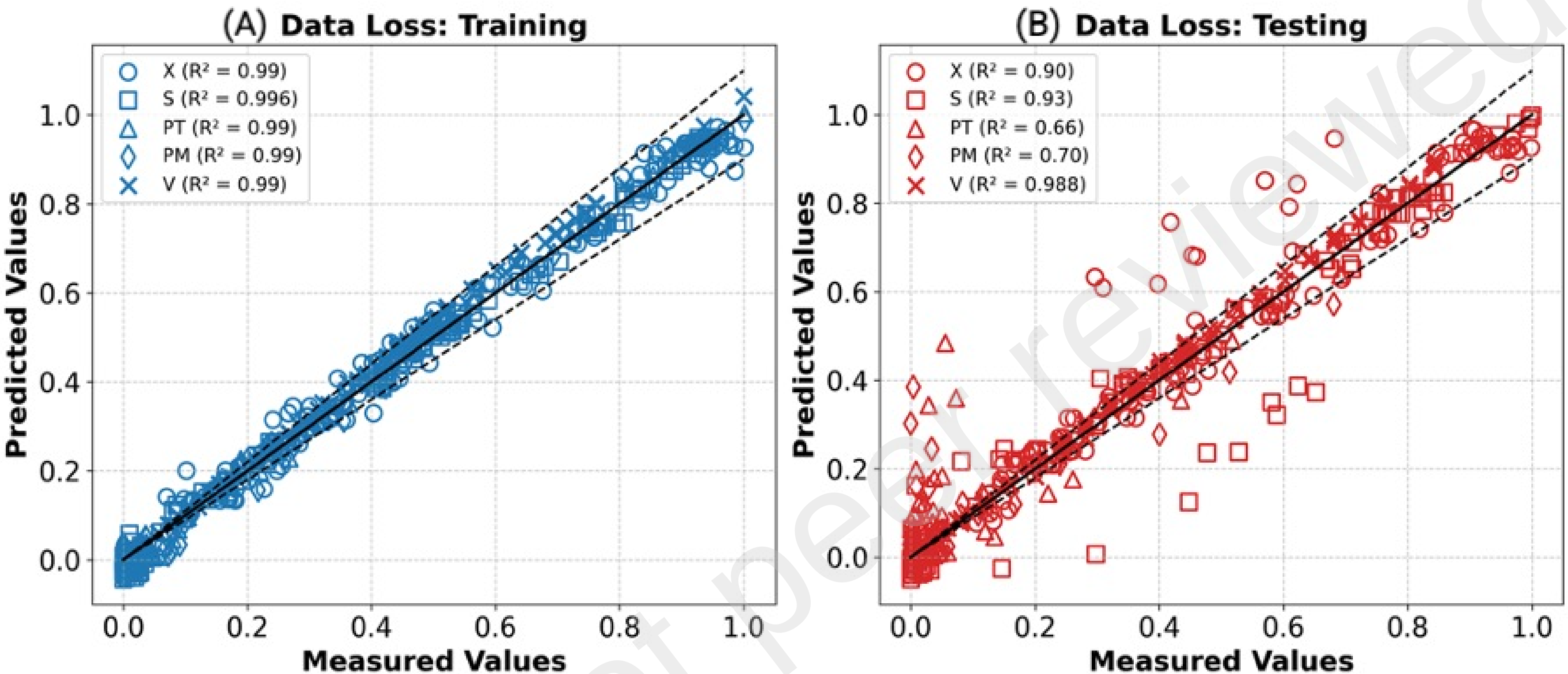


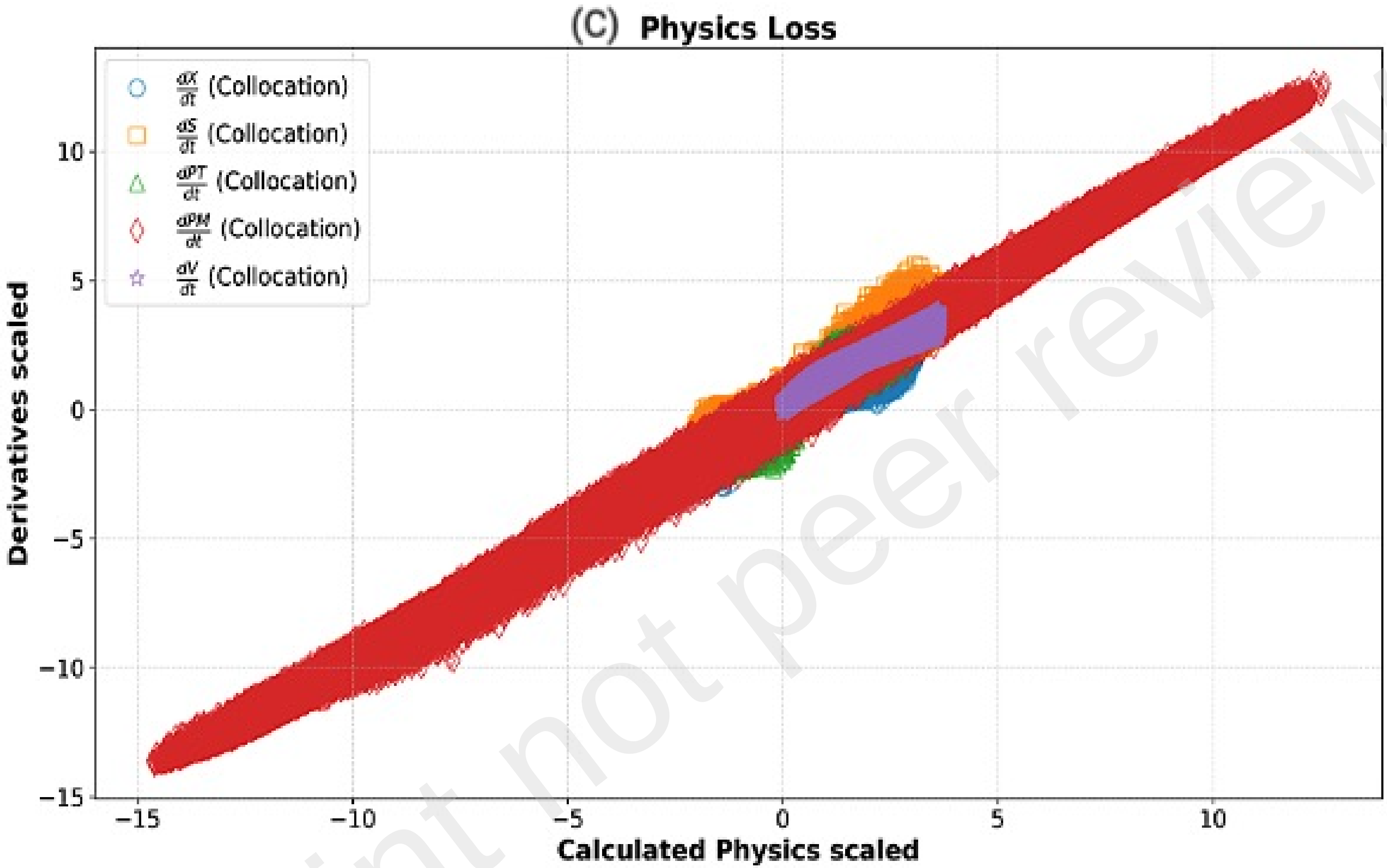
(A) Data Loss: Training
Biomass (R² = 0.995)
Volume (R² = 0.979)
Predicted Values
Measured Values
0.0
0.2
0.4
0.6
0.8
1.0
(B) Data Loss: Testing
Biomass (R² = 0.991)
Volume (R² = 0.985)
Predicted Values
Measured Values

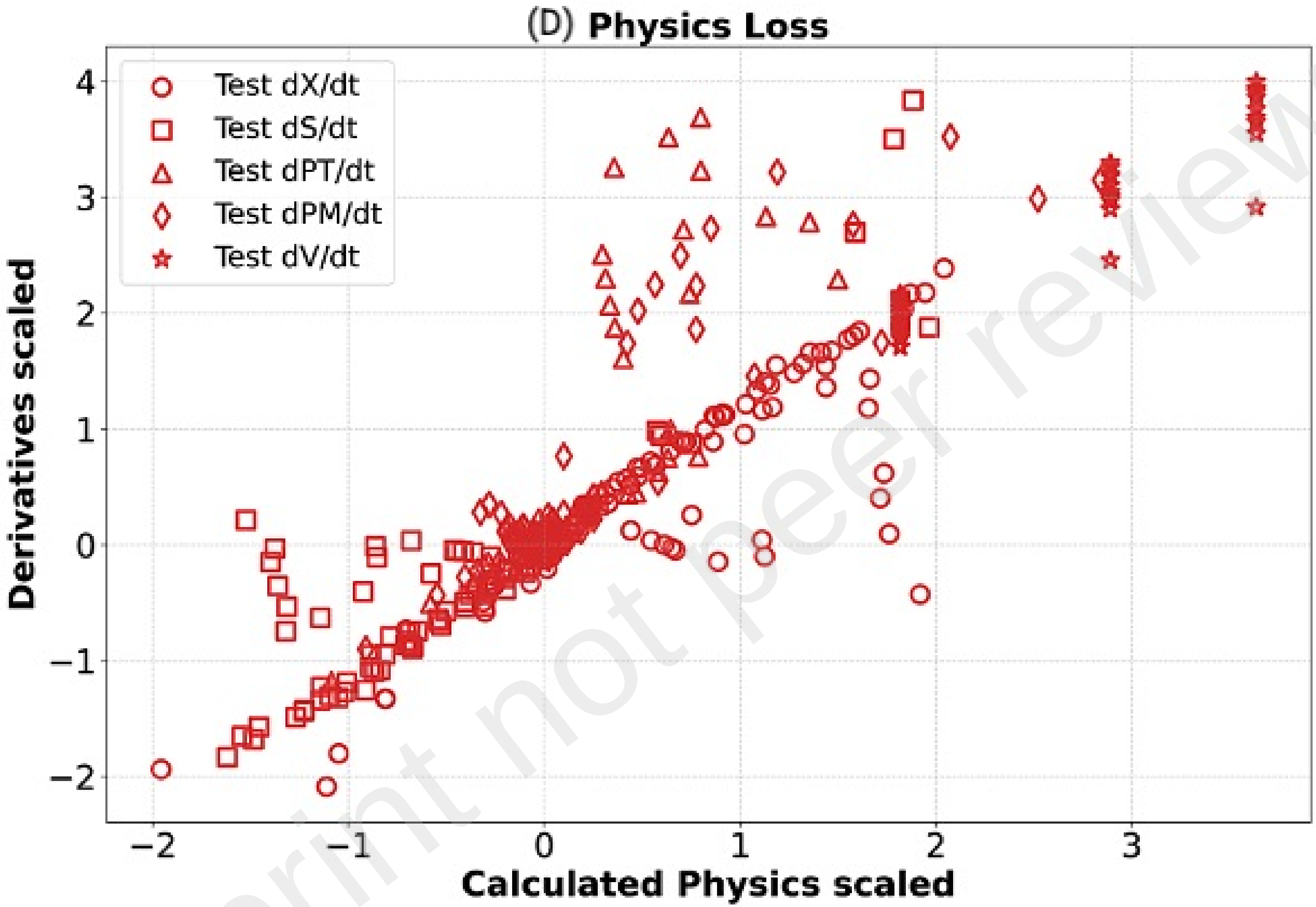

(C) Physics Loss
$\frac{dX}{dt}$ (Collocation)
$\frac{dV}{dt}$ (Collocation)
Derivatives scaled
Calculated Physics scaled
5
0
−5
−10
−15
−20
−25
−20
−15
−10
−5
0
5

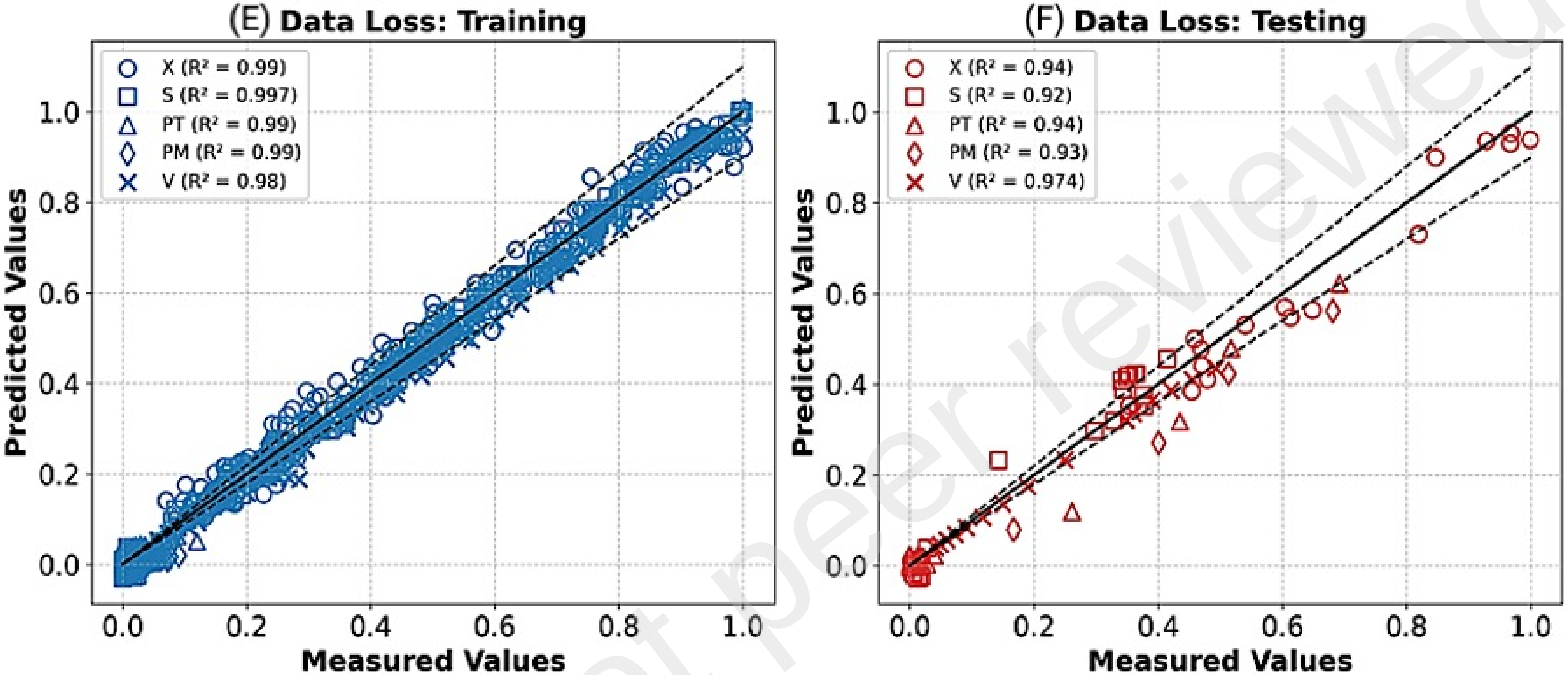


(D) Physics Loss
Test Biomass (dX/dt)
Test Volume (dV/dt)
Derivatives scaled
Calculated Physics scaled
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

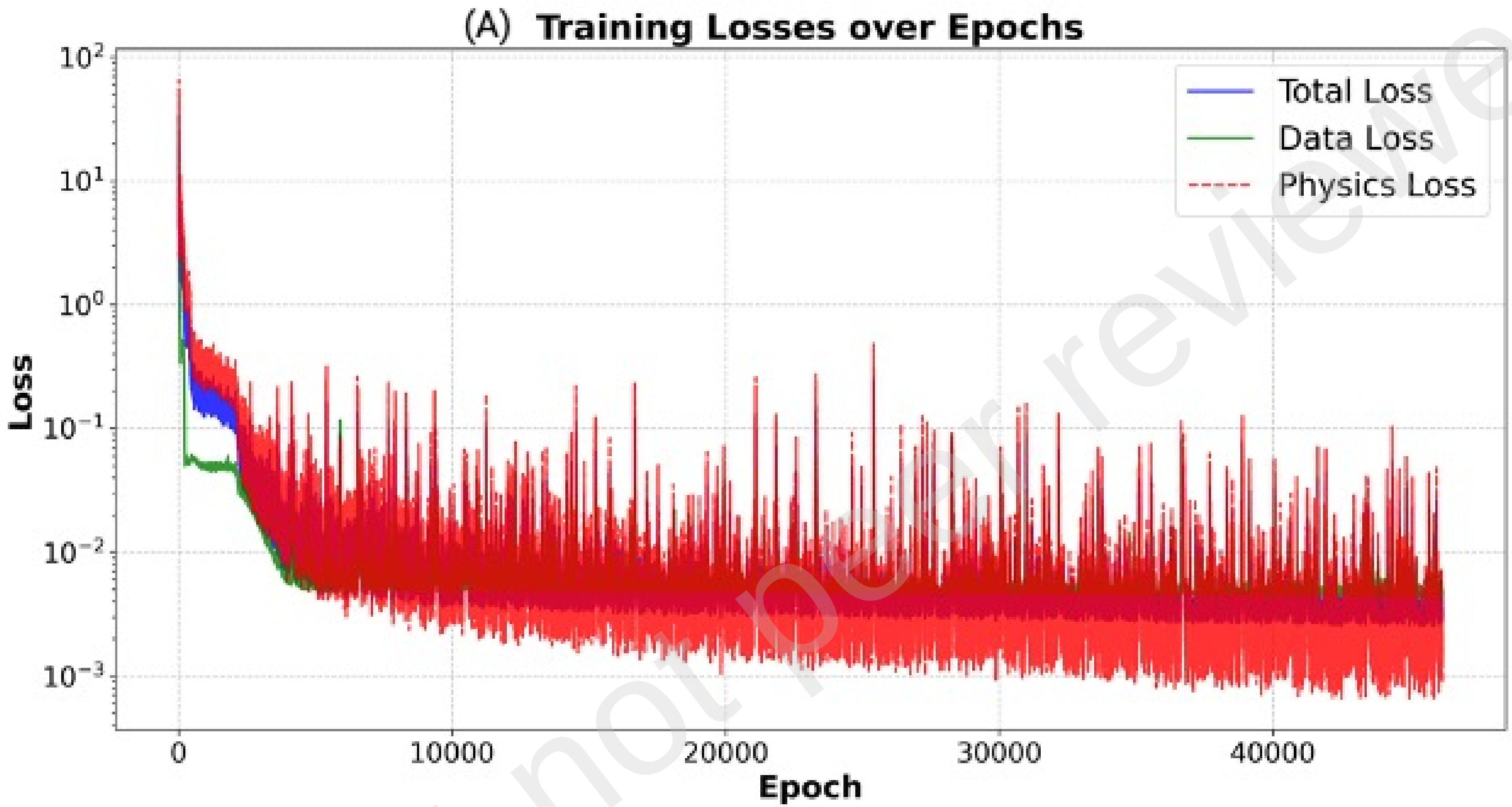

(A) Data Loss: Training
X (R² = 0.99)
S (R² = 0.996)
PT (R² = 0.99)
PM (R² = 0.99)
V (R² = 0.99)
Predicted Values
Measured Values
(B) Data Loss: Testing
X (R² = 0.90)
S (R² = 0.93)
PT (R² = 0.66)
PM (R² = 0.70)
V (R² = 0.988)
Predicted Values
Measured Values

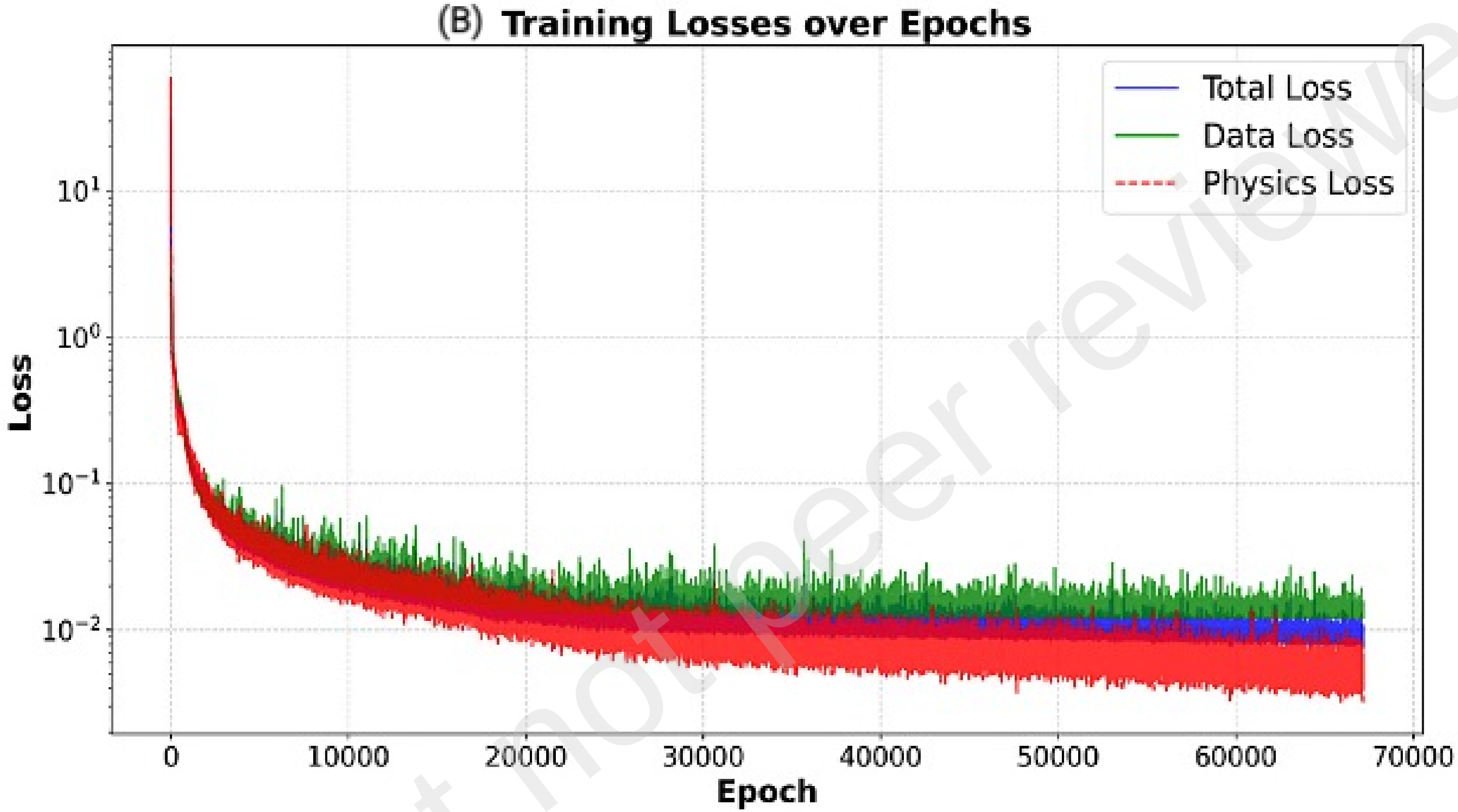

(C) Physics Loss
$\frac{dX}{dt}$ (Collocation)
$\frac{dS}{dt}$ (Collocation)
$\frac{dPT}{dt}$ (Collocation)
$\frac{dPM}{dt}$ (Collocation)
$\frac{dV}{dt}$ (Collocation)
Derivatives scaled
Calculated Physics scaled
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10


(D) Physics Loss
Test dX/dt
Test dS/dt
Test dPT/dt
Test dPM/dt
Test dV/dt
Derivatives scaled
Calculated Physics scaled
4
3
2
1
0
−1
−2
−2
−1
0
1
2
3

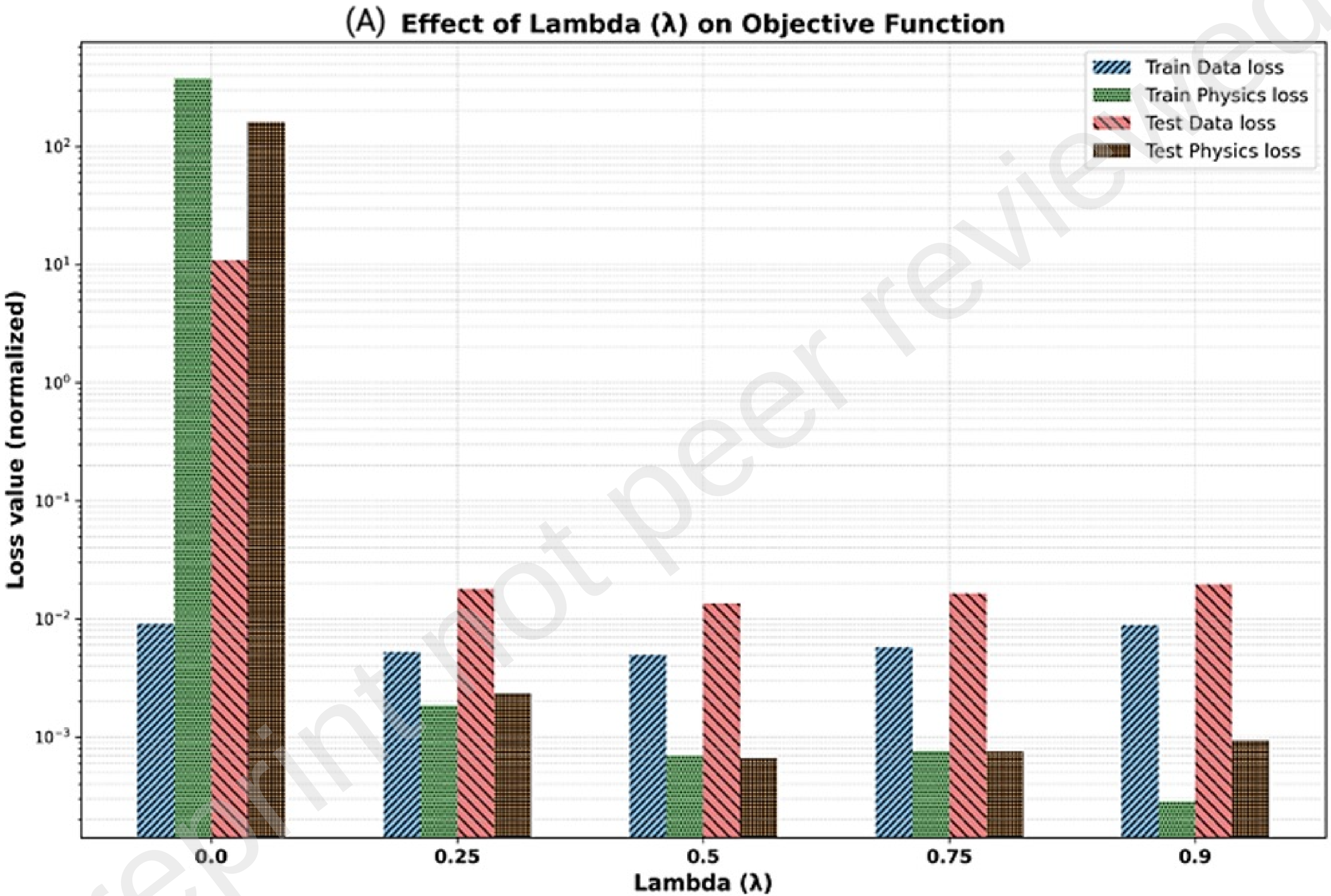


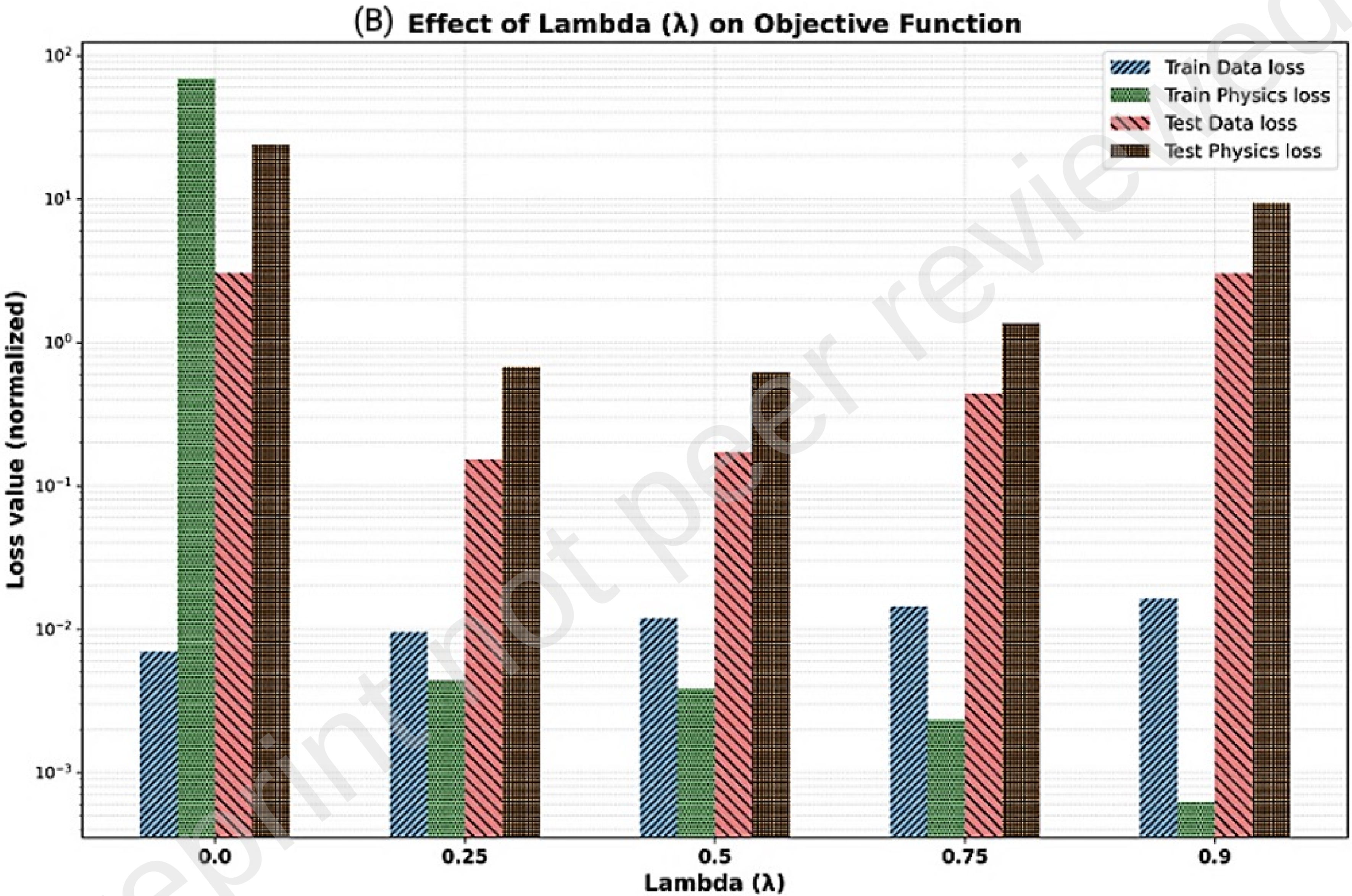
(E) Data Loss: Training
X (R² = 0.99)
S (R² = 0.997)
PT (R² = 0.99)
PM (R² = 0.99)
V (R² = 0.98)
Predicted Values
Measured Values
(F) Data Loss: Testing
X (R² = 0.94)
S (R² = 0.92)
PT (R² = 0.94)
PM (R² = 0.93)
V (R² = 0.974)
Predicted Values
Measured Values


(A) Training Losses over Epochs
Total Loss
Data Loss
Physics Loss
Loss
Epoch
0
10000
20000
30000
40000
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}

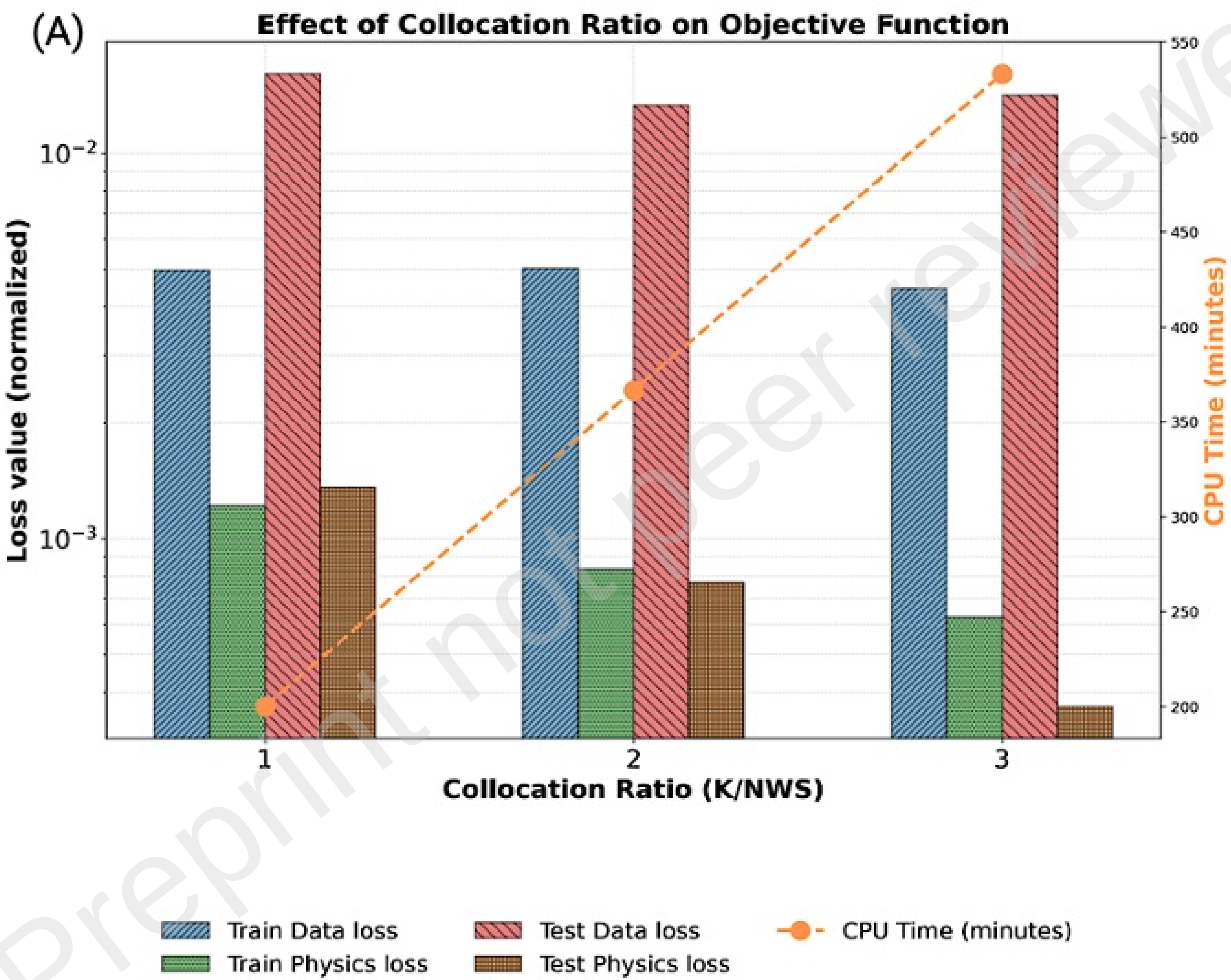


(B) Training Losses over Epochs
Total Loss
Data Loss
Physics Loss
Loss
10^1
10^0
10^{-1}
10^{-2}
0
10000
20000
30000
40000
50000
60000
70000
Epoch


(A) Effect of Lambda (λ) on Objective Function
Train Data loss
Train Physics loss
Test Data loss
Test Physics loss
Loss value (normalized)
10²
10¹
10⁰
10⁻¹
10⁻²
10⁻³
0.0
0.25
0.5
0.75
0.9
Lambda (λ)

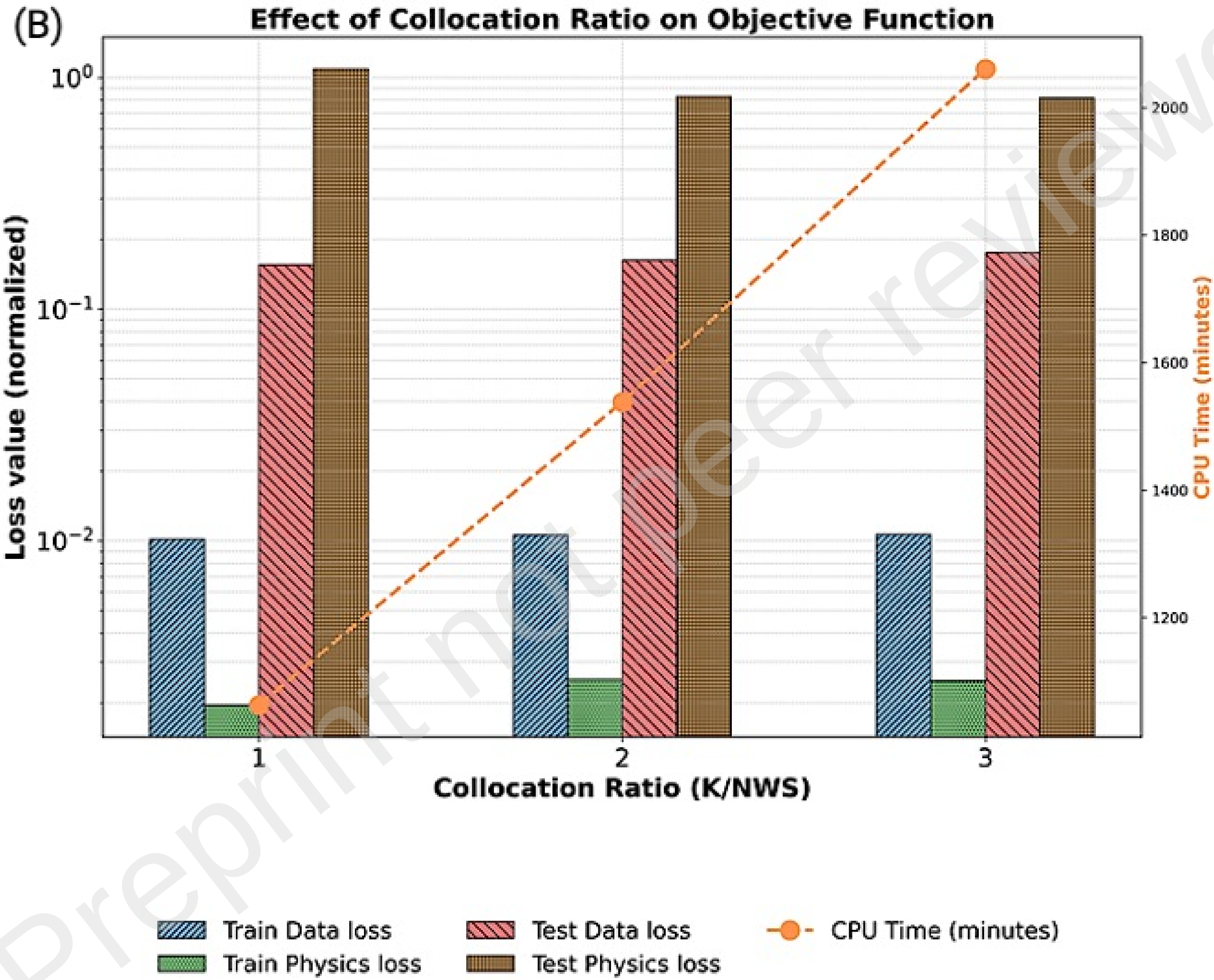


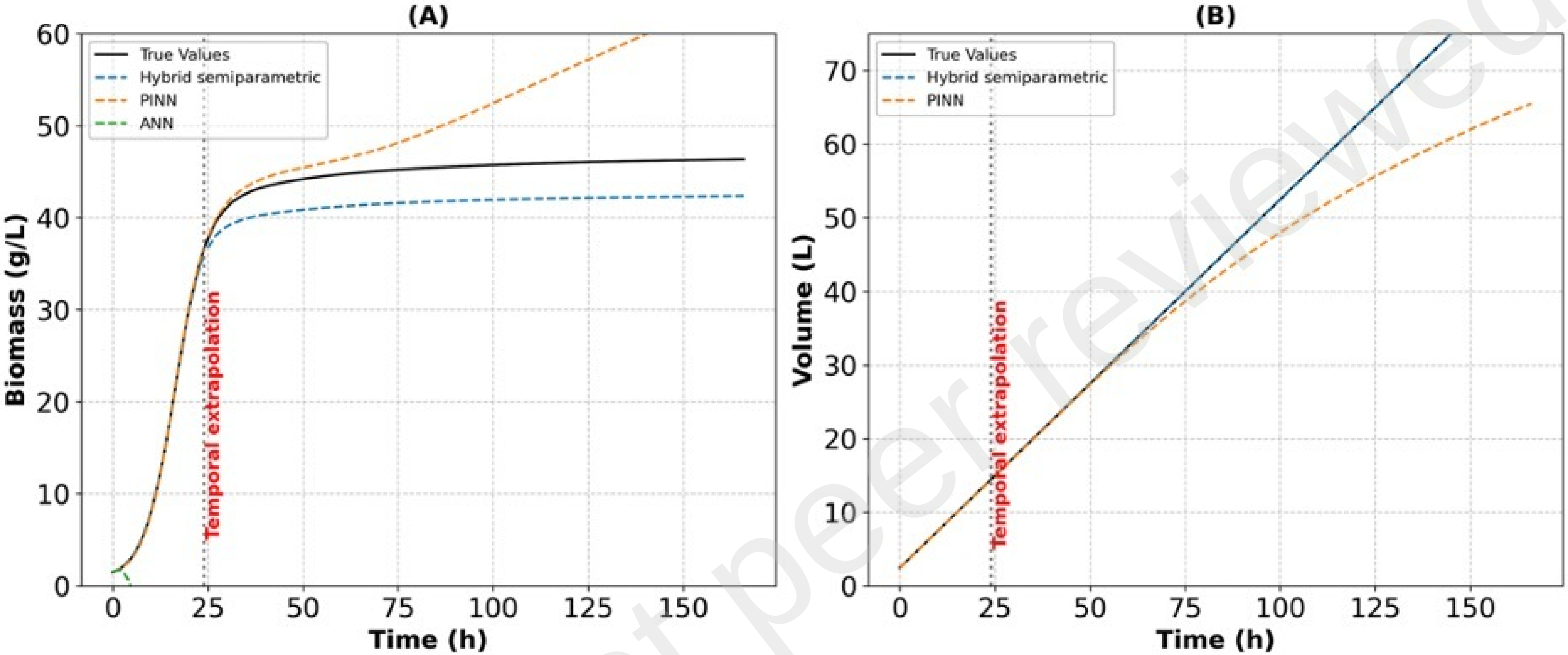
(B) Effect of Lambda (λ) on Objective Function
Train Data loss
Train Physics loss
Test Data loss
Test Physics loss
Loss value (normalized)
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0.0
0.25
0.5
0.75
0.9
Lambda (λ)

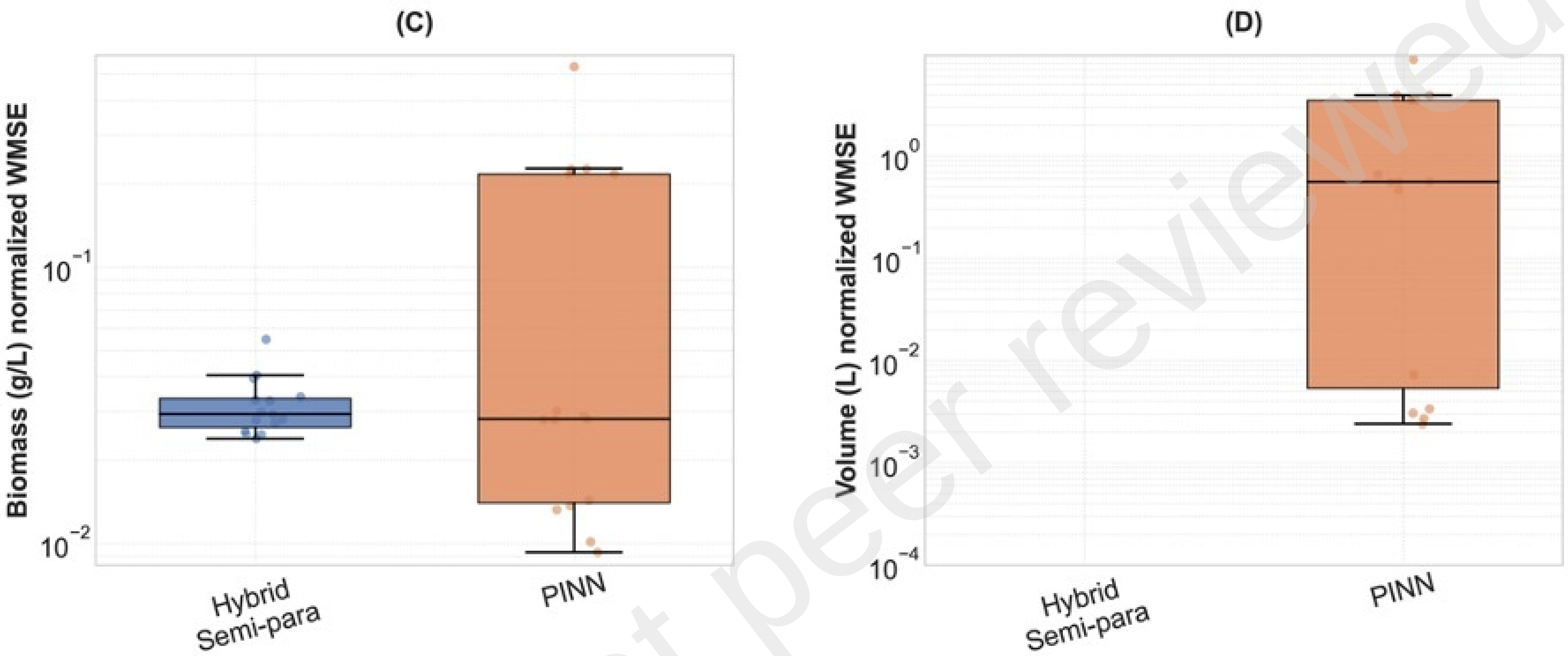


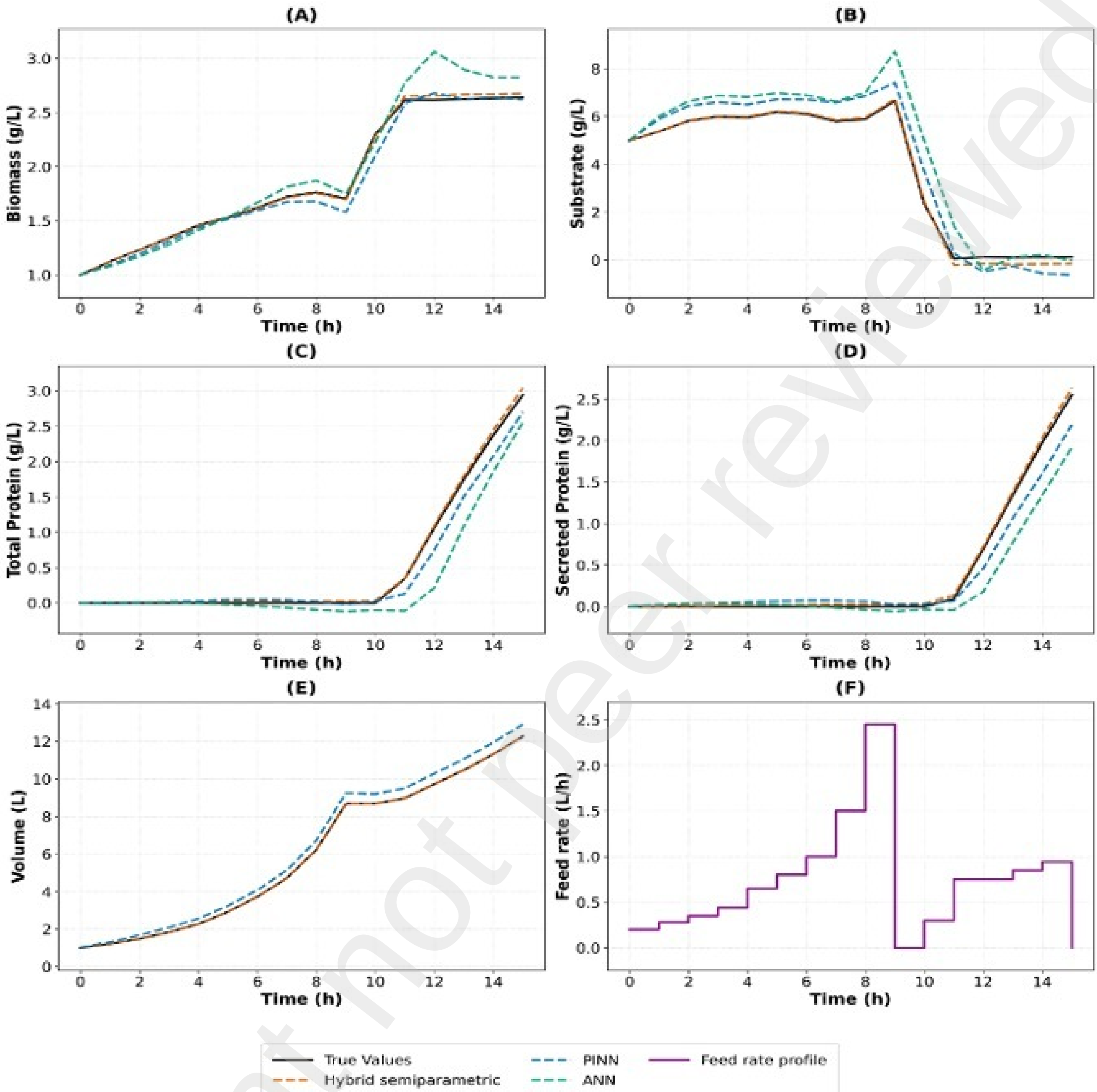
(A)
Effect of Collocation Ratio on Objective Function
Loss value (normalized)
10^{-2}
10^{-3}
CPU Time (minutes)
550
500
450
400
350
300
250
200
1
2
3
Collocation Ratio (K/NWS)
Train Data loss
Train Physics loss
Test Data loss
Test Physics loss
CPU Time (minutes)

(B)


Effect of Collocation Ratio on Objective Function
Loss value (normalized)
10^0
10^{-1}
10^{-2}
CPU Time (minutes)
2000
1800
1600
1400
1200
1
2
3
Collocation Ratio (K/NWS)
Train Data loss
Train Physics loss
Test Data loss
Test Physics loss
CPU Time (minutes)


(A)
(B)
True Values
Hybrid semiparametric
PINN
ANN
Biomass (g/L)
Volume (L)
Time (h)
Temporal extrapolation


(C)
Biomass (g/L) normalized WMSE
10^{-1}
10^{-2}
Hybrid Semi-para
PINN
(D)
Volume (L) normalized WMSE
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Hybrid Semi-para
PINN



(A)
Biomass (g/L)
Time (h)
(B)
Substrate (g/L)
Time (h)
(C)
Total Protein (g/L)
Time (h)
(D)
Secreted Protein (g/L)
Time (h)
(E)
Volume (L)
Time (h)
(F)
Feed rate (L/h)
Time (h)
True Values
Hybrid semiparametric
PINN
ANN
Feed rate profile